

Balanced Randomization for Efficient Approximation of Exact Permutation Tests

Sudesh Srivastav*

Oumar Thiero†

(Short title: *Balanced Randomization for Permutation Tests*)

Abstract

Resampling methods, such as Monte Carlo permutation and bootstrap procedures, are central to nonparametric inference; however, their p -values exhibit inherent simulation variability. We propose a deterministic alternative, the BR test, which utilizes balanced incomplete block designs (BIBDs) to select a structured subset of treatment allocations. Conditional on a chosen design, the method yields fully reproducible p -values by removing Monte Carlo randomness. We show that, for linear allocation statistics, BR supports satisfying exact first- and second-order balance preserve the corresponding mean and variance of the full allocation distribution. We further establish asymptotic validity under sequences of near-balanced designs whose inclusion probabilities converge to those of the uniform allocation distribution. Simulation experiments demonstrate that Balanced Randomization can approximate exact permutation p -values accurately in the settings considered here, often with fewer statistic evaluations than stochastic resampling methods. A high-dimensional illustration demonstrates that the method retains close agreement with permutation-based benchmarks while improving reproducibility in large-scale permutation testing problems.

Keywords: permutation test; balanced incomplete block design; design-based resampling; inclusion probabilities; reproducibility; nonparametric inference

1 Introduction

Nonparametric permutation tests are widely used when parametric modeling assumptions are questionable, because under exchangeability they provide exact Type I error control without requiring distributional normality (4; 14). They are particularly attractive in small samples and under skewed or otherwise non-Gaussian distributions, where asymptotic approximations may be unreliable.

Resampling methods generate pseudo-samples or reallocations from the observed data in order to approximate reference distributions that are analytically intractable (11). In the balanced two-sample setting considered here, common approaches include the full permutation (FP) test, which enumerates all $\binom{N}{n}$ distinct treatment allocations; Monte Carlo permutation (MC) tests based on S_{MC} randomly drawn allocations; and standard nonparametric bootstrap (BA) procedures with S_{BA} resamples. Although FP provides exact randomization inference, exhaustive enumeration becomes computationally prohibitive once the number of allocations is moderately large. MC and BA methods are often practical alternatives, but because they rely on random resampling, their p -values

*Department of Biostatistics and Data Science, School of Public Health and Tropical Medicine, Tulane University, New Orleans, LA, USA. ssrivas@tulane.edu

†Department of Public Health, Faculty of Medicine and Stomatology, University of Sciences, Technics and Technologies of Bamako, Bamako, Mali. outhiero@gmail.com

may exhibit noticeable Monte Carlo variability unless the number of resamples is sufficiently large (15). More broadly, modern analyses of resampling procedures emphasize that the validity and interpretation of bootstrap and permutation methods depend on the structure of the null hypothesis, the resampling mechanism, and the statistic under consideration (10; 7).

The nonparametric bootstrap targets a sampling distribution under independent and identically distributed assumptions rather than the exact permutation distribution arising under exchangeability (3). Nevertheless, bootstrap procedures are widely used in practice as general-purpose nonparametric tools, and we include BA as a computational comparator in order to assess the practical behavior of the proposed Balanced Randomization method relative to commonly used resampling approaches. Accordingly, BA is not treated here as a direct approximation to the exact permutation distribution, but rather as a widely used stochastic resampling benchmark for practical comparison. This distinction is important: BA targets an i.i.d. sampling distribution under model-based assumptions, whereas FP and BR target the finite-sample permutation distribution under exchangeability.

Throughout, FP denotes a high-precision permutation benchmark that targets the exact permutation distribution under exchangeability by drawing a large number of allocations uniformly from $\mathcal{A}_{N,n}$. In the simulation study, we implement FP via Monte Carlo sampling with a large number of draws (up to $S_{\text{FP}} = 10^6$ per dataset) rather than literal enumeration whenever $\binom{N}{n}$ is prohibitively large, which yields an accurate approximation to the full permutation distribution at the reported precision. MC denotes a standard Monte Carlo approximation based on a smaller sample S_{MC} (here $S_{\text{MC}} = 10^4$), and BA denotes a nonparametric bootstrap procedure with $S_{\text{BA}} = 10^4$ resamples drawn by independent resampling within groups. It is important to emphasize that FP (and the related benchmarks used below) target the finite-sample permutation distribution under exchangeability, whereas BA targets the i.i.d. sampling distribution obtained by resampling within groups. Accordingly, BA is included as a widely used nonparametric comparator rather than as a direct approximation to the permutation distribution.

This work introduces a *Balanced Randomization* (BR) method that leverages combinatorial design theory—specifically balanced incomplete block designs (BIBDs) and related group actions—to construct a structured, deterministic subset of the complete allocation space. The guiding idea is to preserve low-order inclusion behavior while replacing random resampling by a fixed support of allocations. Conditional on the chosen design, BR removes Monte Carlo randomness and yields deterministic, fully reproducible p -values. In contrast to variance-reduction strategies that still rely on stochastic sampling (1; 5), BR is deterministic by construction.

Recent work has also explored structured and variance-reduced approximations to permutation inference, including random-permutation tests, sequential Monte Carlo procedures, and exact randomization methods under restricted group actions (1; 2; 7). The present BR framework is closest in spirit to these approaches, but replaces stochastic sampling by fixed, carefully balanced supports.

Scope. Throughout this paper, we restrict attention to permutation-based inference under null hypotheses for which the observations are exchangeable. The proposed method is intended as a deterministic, design-based approximation to exact randomization tests. While we leverage exact BIBDs where feasible, we also consider near-balanced designs in settings where exact constructions are unavailable, thereby broadening the practical applicability of the framework. Our primary focus is on small to moderate sample sizes, where Monte Carlo variability can materially affect numerical inference and where structured allocation schemes may offer meaningful gains in reproducibility and efficiency.

The main contributions are threefold: (i) we introduce a deterministic resampling scheme based

on structured subsets of the allocation space motivated by BIBDs and related group actions; (ii) we establish absence of Monte Carlo randomness, exact low-order moment matching for linear allocation statistics under suitable balance conditions, and a sufficient condition for asymptotic validity under sequences of near-balanced supports; and (iii) we provide simulation and application evidence showing that BR can closely approximate permutation benchmarks while improving reproducibility and reducing computational burden in the settings considered here. This perspective also complements recent developments in permutation methodology for high-dimensional problems, where careful construction of valid resampling schemes remains central to reliable finite-sample inference (6). Method definition, standing assumptions, and the principal theoretical and empirical findings are presented in the main text, while design diagnostics, extended runtime summaries, and supplementary high-dimensional comparisons are collected in Appendix C.

2 Combinatorial Design Motivation

For balanced two-sample permutation problems, assigning N observations into groups of size n is equivalent to selecting one treatment group from the collection

$$\mathcal{A}_{N,n} = \{A \subseteq \{1, \dots, N\} : |A| = n\},$$

whose cardinality is $\binom{N}{n}$. This allocation-space representation is the natural combinatorial object for the BR construction. The full collection of such subsets satisfies the same counting identities as a balanced incomplete block design (BIBD) with parameters

$$(v, b, r, k, \lambda) = \left(N, \binom{N}{n}, \binom{N-1}{n-1}, n, \binom{N-2}{n-2} \right),$$

which obey the standard relations

$$vr = bk, \quad \lambda(v-1) = r(k-1).$$

Here v denotes the number of elements, b the number of blocks, k the block size, r the replication number, and λ the pairwise concurrence parameter. We use this correspondence as a conceptual device to motivate the construction of structured, approximately balanced subsets of the full allocation space; we do not claim that the complete allocation collection should be regarded as a nontrivial BIBD object in the usual design-theoretic sense. Our use of BIBD terminology is therefore constructive rather than ontological: the purpose is to exploit balance identities to build deterministic supports with desirable low-order inclusion properties, not to reclassify the full permutation space as a design object of independent interest.

2.1 Balanced Randomization construction

The following procedure summarizes the construction of the Balanced Randomization (BR) support.

1. **Input.** Total sample size N and group size n .
2. **Feasibility check.** Determine whether an exact $2-(N, n, \lambda)$ BIBD is plausible by checking the necessary arithmetic conditions

$$Nr = bn, \quad (N-1)\lambda = r(n-1).$$

These conditions are necessary but not sufficient for the existence of an exact design.

3. **Base construction.** If an exact construction is available, form a base design

$$\mathcal{D} = \{B_1, \dots, B_b\} \subseteq \mathcal{A}_{N,n}$$

using known algebraic or cyclic constructions. Otherwise, construct a near-balanced collection

$$\mathcal{D} = \{B_1, \dots, B_{b'}\} \subseteq \mathcal{A}_{N,n}$$

using a discrete optimization heuristic designed to approximately match first- and second-order inclusion behavior. More specifically, when an exact construction is unavailable, the near-balanced support is chosen to minimize a low-order discrepancy criterion of the form

$$D(\mathcal{D}) = w_1 \sum_{i=1}^N (\hat{\pi}_i(\mathcal{D}) - \pi_i^*)^2 + w_2 \sum_{1 \leq i < j \leq N} (\hat{\pi}_{ij}(\mathcal{D}) - \pi_{ij}^*)^2,$$

where

$$\pi_i^* = \frac{n}{N}, \quad \pi_{ij}^* = \frac{n(n-1)}{N(N-1)}$$

are the first- and second-order inclusion probabilities under the full permutation measure. In practice, the heuristic proceeds by iterative exchanges or swaps among candidate allocations so as to reduce $D(\mathcal{D})$ and thereby improve agreement with the target inclusion structure of the full allocation distribution.

4. **Optional group expansion.** If the base design admits a nontrivial group action $\mathcal{G}$ (for example, cyclic shifts or other structured permutations of the indices) on $\{1, \dots, N\}$, define

$$\mathcal{B}_R = \text{Unique}\{g(B_i) : g \in \mathcal{G}, i = 1, \dots, |\mathcal{D}|\},$$

where duplicate allocations are removed. Otherwise, set $\mathcal{B}_R = \mathcal{D}$.

5. **Inference.** Evaluate the test statistic $T(A)$ for each allocation $A \in \mathcal{B}_R$ and compute the BR p -value as the empirical tail probability over the fixed support. For a two-sided test,

$$p_{\text{BR}} = \frac{1}{|\mathcal{B}_R|} \sum_{A \in \mathcal{B}_R} I(|T(A)| \geq |T_{\text{obs}}|),$$

where T_{obs} is the observed test statistic.

Figure 1 provides a schematic overview of the construction. The optional group-expansion step need not increase the final support size, since distinct group actions may generate duplicate allocations.

Remark 1 (Counting motivation only). Although the full collection $\mathcal{A}_{N,n}$ satisfies the same counting identities as a 2 -(N, n, λ) design, permutation-based inference in the present setting is most naturally indexed by distinct treatment allocations rather than by all ordered permutations of the labels. The BIBD analogy is therefore used only to motivate structured subsets of the allocation space.

Remark 2 (Scope of BIBD analogy). The use of BIBD terminology in this paper is limited to first- and second-order inclusion behavior. We do not rely on higher-order balance properties of classical designs, nor do we require that the BR supports form exact t -designs for $t > 2$. The key quantities for our purposes are the marginal and pairwise inclusion probabilities of allocation indicators, which are precisely the objects controlled by 2 -(v, k, λ) balance.

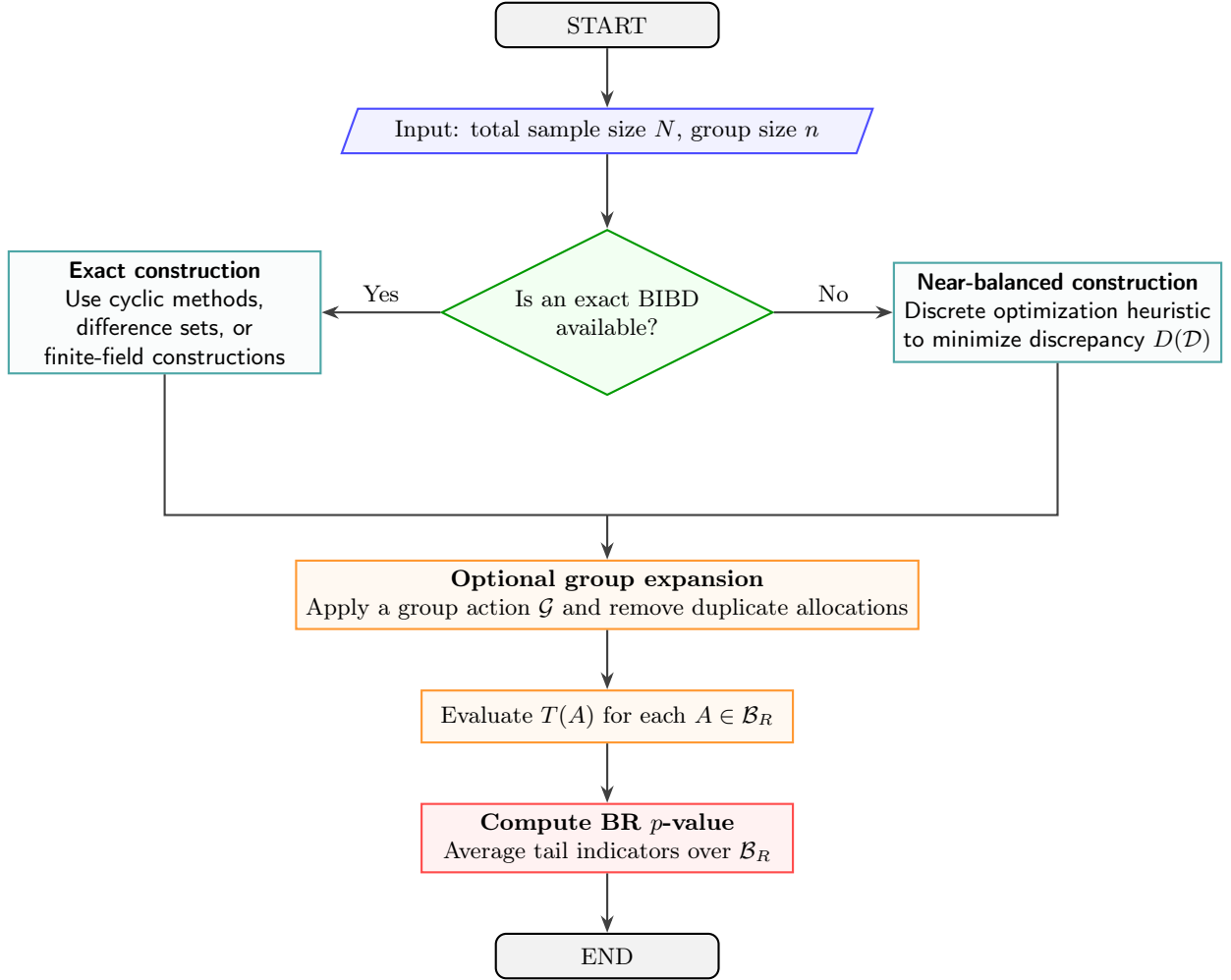


Figure 1: Schematic overview of the Balanced Randomization (BR) construction. The procedure branches according to whether an exact design is available, then optionally enlarges the support through a group action before test-statistic evaluation.

2.2 Explicit BIBD construction

To make the combinatorial idea concrete, we give a standard example of a BIBD that can serve as a BR support.

Worked example ($N = 7, n = 3$). Let the observation indices be

$$\{1, 2, 3, 4, 5, 6, 7\}.$$

A classical 2-(7, 3, 1) design is given by the seven blocks

$$\begin{aligned} B_1 &= \{1, 2, 4\}, & B_5 &= \{2, 5, 7\}, \\ B_2 &= \{1, 3, 5\}, & B_6 &= \{3, 4, 7\}, \\ B_3 &= \{1, 6, 7\}, & B_7 &= \{4, 5, 6\}, \\ B_4 &= \{2, 3, 6\}. \end{aligned}$$

This design has

$$v = 7, \quad k = 3, \quad b = 7, \quad r = 3, \quad \lambda = 1.$$

Hence each element appears in exactly three blocks, and each pair of distinct elements appears together in exactly one block. The standard BIBD identities

$$vr = bk, \quad \lambda(v - 1) = r(k - 1)$$

are immediately verified.

Interpretation. Each block represents one allocation of size $n = 3$. Instead of using all $\binom{7}{3} = 35$ possible allocations, the design uses only $b = 7$ carefully selected allocations while preserving exact first- and second-order balance. This illustrates the basic BR idea: use a much smaller structured support that still reflects the low-order combinatorial features of the complete allocation space.

2.3 Worked example: orbit structure and seed choice for $N = 7, n = 3$

To illustrate how Balanced Randomization (BR) depends on the underlying design, we consider a fully worked example with $N = 7$ observations, group size $n = 3$, and the 2-(7, 3, 1) design from the preceding example. The index set is represented on the cyclic group $\mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$, and we fix the observation vector

$$X = (x_0, x_1, x_2, x_3, x_4, x_5, x_6) = (15, 10, 12, 22, 13, 14, 11).$$

For any allocation $A \subset \{0, \dots, 6\}$ of size $|A| = 3$, the test statistic is the absolute difference in group means,

$$T(A) = \left| \frac{1}{3} \sum_{i \in A} x_i - \frac{1}{4} \sum_{j \notin A} x_j \right|.$$

The exact randomization benchmark is obtained by evaluating $T(A)$ over all $\binom{7}{3} = 35$ distinct allocations in $\mathcal{A}_{7,3}$; in this example, the resulting full-permutation p-value is

$$p_{\text{FP}} = \frac{11}{35} \approx 0.3143.$$

Let g denote the generator of the cyclic group, $g(i) = i + 1 \pmod{7}$, and write $g^k(A)$ for the k -fold image of A under g . For illustration, we take the observed allocation to be the seed allocation, $A_{\text{obs}} = A_{\text{seed}}$. The BR orbit is

$$\mathcal{B}(A_{\text{seed}}) = \{g^k(A_{\text{seed}}) : k = 0, \dots, 6\},$$

and the corresponding BR p -value is

$$p_{\text{BR}}(A_{\text{seed}}) = \frac{1}{7} \sum_{k=0}^6 \mathbf{1}\{|T(g^k(A_{\text{seed}}))| \geq |T(A_{\text{obs}})|\}.$$

Difference-set seed and exact moment matching. Using the cyclic difference-set block $A_1 = \{1, 2, 4\}$ from the 2-(7, 3, 1) design as the seed, the orbit

$$\mathcal{B}(A_1) = \{\{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 6\}, \{0, 4, 5\}, \{1, 5, 6\}, \{0, 2, 6\}, \{0, 1, 3\}\}$$

forms the complete cyclic 2-(7, 3, 1) design. For the signed linear contrast

$$L(A) = \frac{1}{3} \sum_{i \in A} x_i - \frac{1}{4} \sum_{j \notin A} x_j,$$

the induced first- and second-order inclusion probabilities match those of the full allocation distribution exactly. Hence Proposition 1 implies that the BR mean and variance of $L(A)$ coincide with their full-allocation counterparts.

For the two-sided statistic $T(A) = |L(A)|$, exact moment matching is not guaranteed by Proposition 1, since the absolute-value transformation is nonlinear. Nevertheless, the resulting BR tail probability provides a deterministic approximation to the full permutation tail probability. In this example, the full-permutation p -value is

$$p_{\text{FP}} = \frac{11}{35} \approx 0.3143,$$

while the BR value based on the cyclic difference-set support is

$$p_{\text{BR}}(A_1) = \frac{2}{7} \approx 0.2857.$$

Thus the well-balanced cyclic seed yields a two-sided BR p -value close to the full allocation benchmark, while exact moment matching should be interpreted for the underlying signed linear contrast $L(A)$, not for the nonlinear statistic $|L(A)|$.

Unbalanced seeds and variance distortion. If, instead of the difference-set seed, we initialize BR with other blocks from the same $N = 7$ allocation space, the resulting orbits no longer satisfy the 2-(7, 3, 1) inclusion structure. For example, a “single-gap” cluster such as $A_3 = \{0, 1, 6\}$ or a more asymmetric block such as $A_2 = \{1, 3, 5\}$ generates orbits whose pairwise co-occurrence frequencies deviate from the full allocation probabilities. In these cases, the BR mean and variance of the signed linear contrast $L(A)$ need not match the corresponding full-allocation values, and the induced two-sided BR p -values based on $T(A) = |L(A)|$ can be substantially larger or smaller than p_{FP} .

A representative summary for this $N = 7$ example is given in Table 14 in Appendix 8, which groups seeds into three combinatorial classes (cyclic difference-set, asymmetric sub-orbit, and single-gap cluster) and reports the resulting BR moments and p -values. Only the cyclic difference-set seed

reproduces the full-allocation mean and variance of the underlying signed linear contrast $L(A)$ exactly, while the agreement observed for the nonlinear statistic $T(A) = |L(A)|$ in Table 14 is empirical. Notably, for the specific dataset considered here, all seeds still yield the same testing decision (failure to reject at $\alpha = 0.05$), but the numerical p-values differ appreciably across seeds.

Implication for BR design. This example illustrates two points. First, well-balanced seeds such as cyclic difference sets provide BR supports with exact or near-exact low-order inclusion behavior and consequently accurate approximation to the permutation distribution for linear statistics. Second, arbitrary seeds can induce unbalanced supports with biased variance and shifted reference distributions. The BR framework therefore treats the support as a single carefully constructed exact or near-balanced design for each (N, n) configuration, rather than as an average over multiple seeds.

Remark 3 (Near-balance and computational heuristics). Exact BIBDs exist only for specific parameter combinations, so practical implementation requires a broader class of supports. When no exact construction is available for a given (N, n) , we instead seek a near-balanced support whose inclusion behavior approximates that of the full allocation space. One useful criterion is the discrepancy

$$D(\mathcal{D}) = w_1 \sum_{i=1}^N \left(\hat{\pi}_i(\mathcal{D}) - \frac{n}{N} \right)^2 + w_2 \sum_{1 \leq i < j \leq N} \left(\hat{\pi}_{ij}(\mathcal{D}) - \frac{n(n-1)}{N(N-1)} \right)^2,$$

where $\hat{\pi}_i(\mathcal{D})$ and $\hat{\pi}_{ij}(\mathcal{D})$ denote the empirical first- and second-order inclusion frequencies induced by $\mathcal{D}$. This criterion penalizes deviation from the low-order inclusion structure of the full allocation distribution and motivates the heuristic construction described in Appendix B.

3 Balanced Randomization Test

3.1 Formal definition

For the balanced two-sample problems considered in this paper, the exact randomization distribution is naturally indexed by the allocation space

$$\mathcal{A}_{N,n} = \{A \subseteq \{1, \dots, N\} : |A| = n\},$$

equipped with the uniform distribution over its $\binom{N}{n}$ elements. Let $T(A)$ denote the test statistic evaluated at allocation A , and let T_{obs} be its observed value. The exact permutation p -value is then the corresponding tail probability under the uniform distribution on $\mathcal{A}_{N,n}$.

Definition 1 (Balanced Randomization support). Let

$$\mathcal{D} = \{B_1, \dots, B_b\} \subseteq \mathcal{A}_{N,n}$$

be an exact or near-balanced collection of allocations constructed as in Section 2.1, and let $\mathcal{G}$ be a group acting on $\{1, \dots, N\}$. The *Balanced Randomization* (BR) support is defined by

$$\mathcal{B}_R = \text{Unique}\{g(B_i) : g \in \mathcal{G}, i = 1, \dots, b\} \subseteq \mathcal{A}_{N,n},$$

where duplicate allocations are removed. The induced BR empirical distribution of T is

$$\hat{F}_{BR}(t) = \frac{1}{|\mathcal{B}_R|} \sum_{A \in \mathcal{B}_R} \mathbf{1}\{T(A) \leq t\}.$$

For a two-sided test, the corresponding BR p -value is

$$p_{\text{BR}} = \frac{1}{|\mathcal{B}_R|} \sum_{A \in \mathcal{B}_R} \mathbf{1}\{|T(A)| \geq |T_{\text{obs}}|\}.$$

Because $\mathcal{B}_R$ is finite, the resulting BR p -value is discrete, with resolution $1/|\mathcal{B}_R|$, and is interpreted here as the deterministic empirical tail probability induced by the chosen support. Once the design and group action are fixed, the support $\mathcal{B}_R$ is fixed as well; accordingly, BR is a structured deterministic approximation to the full allocation distribution rather than a stochastic resampling method.

Remark 4 (Finite-sample implementation of permutation benchmarks). In the theoretical development above, the full allocation distribution is defined as the uniform distribution on $\mathcal{A}_{N,n}$, which coincides with the exact permutation distribution under exchangeability. In the simulation and high-dimensional studies, the corresponding permutation benchmarks are implemented via large Monte Carlo samples from $\mathcal{A}_{N,n}$ (with S_{FP} or $S_{\text{MC,bench}}$ up to 10^6), rather than by literal enumeration when $\binom{N}{n}$ is very large. At the reported numerical precision, these large-sample approximations are effectively indistinguishable from the exact permutation distribution in the settings considered here.

Remark 5 (Allocation-space representation). For the balanced two-sample setting considered here, the permutation distribution is equivalently represented by the induced treatment allocations in $\mathcal{A}_{N,n}$, since allocations that differ only by within-group label order yield the same group assignment. Accordingly, the BR construction is formulated on the allocation space rather than on the full ordered permutation group.

3.2 Basic properties

Lemma 1 (Absence of resampling randomness). *Fix observed data X , a design $\mathcal{D}$, and a group action $\mathcal{G}$ generating the BR support $\mathcal{B}_R$. Then the BR p -value is a deterministic function of $(X, \mathcal{D}, \mathcal{G})$ and satisfies*

$$\text{Var}(p_{\text{BR}} \mid X, \mathcal{D}, \mathcal{G}) = 0.$$

Proof. Once $(X, \mathcal{D}, \mathcal{G})$ is fixed, the support $\mathcal{B}_R$ is fixed, the indicator values

$$\mathbf{1}\{|T(A)| \geq |T_{\text{obs}}|\}, \quad A \in \mathcal{B}_R,$$

are all determined, and hence their average is deterministic. $\square$

Remark 6 (Interpretation). Lemma 1 refers only to the absence of Monte Carlo randomness. It does not imply that p_{BR} equals the exact permutation p -value, nor that approximation error is absent. Any discrepancy between BR and the full allocation distribution arises from the restricted support $\mathcal{B}_R \subseteq \mathcal{A}_{N,n}$. In this sense, BR removes the simulation variability associated with random resampling, while any remaining discrepancy is purely due to deterministic approximation of the full allocation distribution.

To study how well BR approximates the exact allocation distribution, it is natural to compare their low-order inclusion behavior. Let

$$Z_i(A) = \mathbf{1}\{i \in A\}, \quad i = 1, \dots, N,$$

denote the allocation indicator for element i .

Proposition 1 (Moment matching for linear allocation statistics). *Consider a statistic of the form*

$$T(A) = \sum_{i=1}^N a_i Z_i(A).$$

If the BR support $\mathcal{B}_R$ has the same first- and second-order inclusion probabilities as the uniform distribution on $\mathcal{A}_{N,n}$, namely

$$P_{\text{BR}}(i \in A) = \frac{n}{N}, \quad P_{\text{BR}}(i, j \in A) = \frac{n(n-1)}{N(N-1)} \quad (i \neq j),$$

then

$$E_{\text{BR}}\{T(A)\} = E\{T(A)\}, \quad \text{Var}_{\text{BR}}\{T(A)\} = \text{Var}\{T(A)\},$$

where the unsubscripted expectation and variance are taken under the uniform distribution on $\mathcal{A}_{N,n}$.

Proof. The expectation of $T(A)$ depends strictly on the first-order inclusion probabilities of the indicators $Z_i(A)$. By linearity of expectation,

$$E\{T(A)\} = E\left\{\sum_{i=1}^N a_i Z_i(A)\right\} = \sum_{i=1}^N a_i E\{Z_i(A)\} = \sum_{i=1}^N a_i \pi_i,$$

where $\pi_i = P(i \in A)$. Similarly, the variance of the linear combination can be expanded as

$$\text{Var}\{T(A)\} = \text{Var}\left(\sum_{i=1}^N a_i Z_i(A)\right) = \sum_{i=1}^N a_i^2 \text{Var}\{Z_i(A)\} + 2 \sum_{1 \leq i < j \leq N} a_i a_j \text{Cov}\{Z_i(A), Z_j(A)\}.$$

Since $Z_i(A)$ is a Bernoulli random variable with success probability π_i , its marginal variance is

$$\text{Var}\{Z_i(A)\} = \pi_i(1 - \pi_i),$$

and for $i \neq j$, the covariance between allocation indicators is

$$\text{Cov}\{Z_i(A), Z_j(A)\} = \pi_{ij} - \pi_i \pi_j,$$

where $\pi_{ij} = P(i, j \in A)$. Thus, both the mean and variance of $T(A)$ are completely determined by the first- and second-order inclusion probabilities $\{\pi_i\}$ and $\{\pi_{ij}\}$. Under the assumptions of the proposition,

$$P_{\text{BR}}(i \in A) = \frac{n}{N}, \quad P_{\text{BR}}(i, j \in A) = \frac{n(n-1)}{N(N-1)},$$

which coincide with the corresponding first- and second-order inclusion probabilities under the uniform allocation distribution on $\mathcal{A}_{N,n}$. Consequently,

$$E_{\text{BR}}\{T(A)\} = E\{T(A)\},$$

and

$$\text{Var}_{\text{BR}}\{T(A)\} = \text{Var}\{T(A)\}.$$

This proves the proposition. □

Remark 7 (Scope of moment matching). Proposition 1 applies directly to linear allocation statistics and to statistics whose leading asymptotic term is linear in the allocation indicators. Although the centering condition $\sum_{i=1}^N a_i = 0$ is not strictly required for the algebraic equality of the first two moments, it is standard and natural for common two-sample contrasts, including differences in sample means or mean ranks. It should be noted that moment matching alone does not imply equality in distribution, but it establishes a rigorous structural baseline for finite-sample approximation.

The preceding moment-matching result shows that exact first- and second-order balance reproduces the mean and variance of linear allocation statistics. We now quantify how deviations from exact second-order balance affect finite-sample variance.

Proposition 2 (Finite-sample variance bound for linear allocation statistics). *Let*

$$T(A) = \sum_{i=1}^N a_i Z_i(A), \quad Z_i(A) = \mathbf{1}\{i \in A\},$$

where Ω denotes the uniform distribution on the full allocation space $\mathcal{A}_{N,n}$. Suppose that the first-order inclusion probabilities under BR match those of the full allocation distribution:

$$\pi_i^{(\text{BR})} = \mathbb{P}_{\text{BR}}(i \in A) = \frac{n}{N}, \quad i = 1, \dots, N.$$

Define

$$\delta_N = \max_{i \neq j} \left| \pi_{ij}^{(\text{BR})} - \frac{n(n-1)}{N(N-1)} \right|,$$

where

$$\pi_{ij}^{(\text{BR})} = \mathbb{P}_{\text{BR}}(i, j \in A).$$

Then

$$|\text{Var}_{\text{BR}}(T) - \text{Var}_{\Omega}(T)| \leq 2\delta_N \sum_{1 \leq i < j \leq N} |a_i a_j|.$$

Equivalently,

$$|\text{Var}_{\text{BR}}(T) - \text{Var}_{\Omega}(T)| \leq C_N \delta_N, \quad C_N := 2 \sum_{1 \leq i < j \leq N} |a_i a_j|.$$

Note that

$$C_N = \left(\sum_{i=1}^N |a_i| \right)^2 - \sum_{i=1}^N a_i^2,$$

so the constant depends only on the magnitude of the coefficient vector. In particular, finite-sample variance distortion is linearly controlled by the magnitude of pairwise imbalance.

Proof. Write

$$T(A) = \sum_{i=1}^N a_i Z_i(A), \quad Z_i(A) = \mathbf{1}\{i \in A\}.$$

Under any probability measure on the allocation space,

$$\text{Var}(T) = \sum_{i=1}^N a_i^2 \text{Var}(Z_i) + 2 \sum_{1 \leq i < j \leq N} a_i a_j \text{Cov}(Z_i, Z_j).$$

Since Z_i is Bernoulli with success probability $\pi_i = \mathbb{P}(i \in A)$,

$$\text{Var}(Z_i) = \pi_i(1 - \pi_i),$$

and for $i \neq j$,

$$\text{Cov}(Z_i, Z_j) = \pi_{ij} - \pi_i\pi_j.$$

Under the full allocation distribution Ω ,

$$\pi_i^{(\Omega)} = \frac{n}{N}, \quad \pi_{ij}^{(\Omega)} = \frac{n(n-1)}{N(N-1)}.$$

By assumption, under BR we also have

$$\pi_i^{(\text{BR})} = \frac{n}{N} \quad \text{for all } i.$$

Therefore the marginal variance terms coincide:

$$\text{Var}_{\text{BR}}(Z_i) = \text{Var}_{\Omega}(Z_i) = \frac{n}{N} \left(1 - \frac{n}{N}\right).$$

Hence the difference between the two variances comes only from the covariance terms:

$$\text{Var}_{\text{BR}}(T) - \text{Var}_{\Omega}(T) = 2 \sum_{1 \leq i < j \leq N} a_i a_j \left(\pi_{ij}^{(\text{BR})} - \pi_{ij}^{(\Omega)} \right).$$

Taking absolute values and using the definition of δ_N ,

$$|\text{Var}_{\text{BR}}(T) - \text{Var}_{\Omega}(T)| \leq 2 \sum_{1 \leq i < j \leq N} |a_i a_j| \left| \pi_{ij}^{(\text{BR})} - \frac{n(n-1)}{N(N-1)} \right| \leq 2\delta_N \sum_{1 \leq i < j \leq N} |a_i a_j|.$$

This proves the result. $\square$

Remark 8. Because Proposition 2 is a self-contained finite-sample result, its full proof is given in the main text rather than deferred to the appendix.

Intuition. For linear allocation statistics, the variance of T is determined by the first- and second-order inclusion behavior of the allocation distribution. The first-order inclusion probabilities govern the marginal variance terms, whereas the second-order inclusion probabilities govern the covariance terms. Because BR matches the first-order inclusion probabilities of the full allocation distribution exactly, the marginal variance contributions coincide under the two schemes. Consequently, the remaining variance distortion is driven entirely by pairwise imbalance, and Proposition 2 shows that this distortion is linearly controlled by δ_N .

Remark 9 (Interpretation). The result shows that, for linear allocation statistics, variance distortion under BR is driven entirely by pairwise imbalance. In particular, exact second-order balance implies exact variance matching.

Remark 10. The bound in Proposition 2 is stated for fixed N , with constant

$$C_N = 2 \sum_{1 \leq i < j \leq N} |a_i a_j|.$$

For asymptotic use over a sequence $N \rightarrow \infty$, a bound of the form

$$|\text{Var}_{\text{BR}}(T) - \text{Var}_{\Omega}(T)| \leq C \delta_N$$

with C independent of N requires an additional uniform restriction on the coefficients, such as

$$\sum_{i=1}^N |a_i| = O(1),$$

which implies $C_N = O(1)$.

Remark 11. The assumption of exact first-order matching is essential for Proposition 2 in its present form. If the marginal inclusion probabilities under BR also differ from those of the full allocation distribution, then additional terms involving

$$\max_{1 \leq i \leq N} \left| \pi_i^{(\text{BR})} - \frac{n}{N} \right|$$

enter the variance comparison.

Remark 12 (Approximate marginal balance). In practice, near-balanced supports may only satisfy

$$\max_{1 \leq i \leq N} \left| \pi_i^{(\text{BR})} - \frac{n}{N} \right| \leq \varepsilon_N$$

for a sequence $\varepsilon_N \rightarrow 0$. In such cases, additional variance terms of order ε_N appear in the comparison. This is consistent with the assumptions of Theorem 1, where both first- and second-order inclusion probabilities for the BR supports are required to converge to their full-allocation counterparts.

3.3 Asymptotic validity

The preceding finite-sample and moment-matching results motivate a corresponding asymptotic statement for sequences of near-balanced supports.

Standing assumptions. Throughout the theoretical developments below, we assume that under the null hypothesis the observations are exchangeable with respect to treatment allocation. For exact low-order moment comparisons, we restrict attention to statistics that are linear in the allocation indicators. For asymptotic validity, we consider sequences of BR supports whose induced allocation measures converge to the uniform allocation distribution in a suitable sense, with low-order inclusion behavior serving as a practical diagnostic for this convergence. In particular, for classical two-sample statistics such as differences in means or mean ranks, combinatorial central limit theorems of Hoeffding (8; 9) provide natural conditions under which Assumption 2 of Theorem 1 holds.

Remark on asymptotic balance. The first- and second-order inclusion conditions used throughout this paper provide practical diagnostics for assessing the quality of a BR support. However, low-order inclusion matching alone does not generally imply convergence of the full BR allocation distribution to the uniform allocation distribution. In particular, matching moments does not, by itself, guarantee weak convergence of distributions. For asymptotic validity, it is therefore natural to require a stronger condition ensuring that the BR allocation measure approximates the uniform allocation measure in an appropriate probabilistic sense.

Theorem 1 (Sufficient condition for asymptotic validity). *Consider a sequence of balanced two-sample problems with $N \rightarrow \infty$ and $n/N \rightarrow \rho \in (0, 1)$. Let P_N denote the uniform allocation measure on $\mathcal{A}_{N,n}$, and let $P_{\text{BR},N}$ denote the empirical allocation measure induced by the BR support $\mathcal{B}_{R,N}$. Suppose the statistic admits the decomposition*

$$T_N(A) = L_N(A) + R_N(A),$$

where

$$L_N(A) = \sum_{i=1}^N a_{Ni} Z_i(A), \quad \sum_{i=1}^N a_{Ni} = 0.$$

Assume that:

1. the observations are exchangeable under H_0 ;
2. the standardized linear part satisfies a combinatorial central limit theorem under P_N ;
3. the remainder $R_N(A)$ is negligible relative to the standard deviation of $L_N(A)$ under P_N ;
4. the BR allocation measure approximates the full allocation measure in total variation:

$$d_{\text{TV}}(P_{\text{BR},N}, P_N) \rightarrow 0.$$

Then the BR and full-allocation distribution functions of the standardized statistic T_N are asymptotically equivalent uniformly in t :

$$\sup_{t \in \mathbb{R}} \left| F_{\text{BR},N}^{(T)}(t) - F_N^{(T)}(t) \right| \rightarrow 0.$$

Consequently, BR critical values and BR p -values are asymptotically valid for T_N .

Remark 13 (Strength of the total variation condition). The total variation condition in Theorem 1 is a sufficient condition rather than a claim automatically guaranteed by first- and second-order balance. It is intentionally strong and is used to state a clean asymptotic validity result. In finite samples, the quality of a BR support is assessed through low-order discrepancy diagnostics and empirical approximation performance.

Proof sketch. Let

$$U_N(A) = \frac{L_N(A) - E_N\{L_N(A)\}}{\sqrt{\text{Var}_N\{L_N(A)\}}},$$

where E_N and Var_N denote expectation and variance under the uniform allocation measure P_N . By assumption (2), U_N converges in distribution to a standard normal random variable under P_N .

Assumption (4) implies that the BR allocation measure and the uniform allocation measure are asymptotically indistinguishable for bounded measurable functions on $\mathcal{A}_{N,n}$. In particular, for every $t \in \mathbb{R}$,

$$|P_{\text{BR},N}\{U_N(A) \leq t\} - P_N\{U_N(A) \leq t\}| \leq d_{\text{TV}}(P_{\text{BR},N}, P_N) \rightarrow 0.$$

Thus, the BR and full-allocation distribution functions of the standardized linear part are asymptotically equivalent. Since the full-allocation distribution of U_N is asymptotically standard normal, the same limiting distribution holds under $P_{\text{BR},N}$.

By assumption (3), the remainder $R_N(A)$ is negligible relative to the standard deviation of $L_N(A)$ under the full allocation measure. The total variation condition in assumption (4) transfers

this negligibility to the BR allocation measure, since events having vanishing probability under P_N also have vanishing probability under $P_{\text{BR},N}$. A Slutsky-type argument therefore establishes the same limiting distribution for the standardized statistic $T_N(A) = L_N(A) + R_N(A)$ under both P_N and $P_{\text{BR},N}$. It follows that

$$\sup_{t \in \mathbb{R}} \left| F_{\text{BR},N}^{(T)}(t) - F_N^{(T)}(t) \right| \rightarrow 0,$$

which guarantees that BR critical values and p -values computed from the deterministic BR support are asymptotically valid for T_N . $\square$

Remark 14 (Interpretation of the theorem). Theorem 1 is intentionally scoped to asymptotically linear statistics whose leading term admits a combinatorial central limit theorem under the full allocation distribution. It should not be read as a universal asymptotic guarantee for arbitrary nonlinear statistics or arbitrary support constructions. Extending the result to broader classes would require additional control of higher-order inclusion behavior and is beyond the scope of the present paper.

Remark 15 (Near-balance in practice). Exact 2 -(N, n, λ) designs exist only for selected parameter combinations. In applications where exact designs are unavailable, BR relies on near-balanced supports constructed to approximate the target inclusion probabilities. The practical performance of these supports depends on design quality, which in this paper is assessed empirically through discrepancy summaries and simulation performance rather than asserted as a universal guarantee.

3.4 Practical interpretation

The theory above supports the following interpretation of BR. First, BR removes Monte Carlo randomness once the support is fixed. Second, exact low-order balance reproduces the mean and variance of linear allocation statistics under the full allocation distribution. Third, near-balanced supports can yield asymptotically valid approximations for asymptotically linear statistics under suitable regularity conditions.

These properties justify BR as a structured deterministic approximation to exact randomization inference. Its main practical appeal is reproducibility: unlike Monte Carlo procedures, repeated runs under the same design return the same reference distribution and the same p -value.

3.5 Procedure

The BR test is implemented through the following deterministic pipeline:

1. Specify the null hypothesis, alternative, and test statistic.
2. Construct an exact or near-balanced support $\mathcal{D} \subseteq \mathcal{A}_{N,n}$ using the methods of Section 2.1.
3. Optionally enlarge the support through a group action and remove duplicate allocations to obtain $\mathcal{B}_R$.
4. Evaluate the test statistic on each allocation in $\mathcal{B}_R$.
5. Compute the BR p -value as the empirical tail probability over this fixed support.

Unlike stochastic resampling methods, the support is constructed once and then reused. Consequently, for a fixed experimental configuration, BR produces the same reference distribution and the same p -value each time it is applied.

Remark 16 (Seed choice and design specificity). The BR framework is design-based: for each (N, n) configuration, inference is conducted conditional on a single, carefully constructed support $\mathcal{D}$ and its group expansion, when used. In particular, we do not average BR p -values over multiple seed blocks or treat the seed choice as an additional randomization device. As the $N = 7$, $n = 3$ example in Section 2.3 illustrates, well-balanced seeds such as cyclic difference sets can reproduce the full allocation mean and variance for linear statistics, whereas arbitrary seeds may induce unbalanced supports with distorted variance and shifted BR reference distributions. Averaging over such heterogeneous seeds would mix well-balanced and poorly balanced designs and can move the resulting p -values further away from the exact permutation benchmark. From this perspective, BR should be interpreted as a deterministic approximation based on a single exact or near-balanced design, not as an average over multiple seed-specific constructions.

Remark 17 (Quadrature interpretation). The BR estimator may be viewed as a deterministic quadrature rule over the permutation distribution, where the support points are selected to match low-order inclusion moments. This perspective emphasizes that BR is a structured approximation to permutation averaging rather than a stochastic resampling estimator.

4 Simulation Study and Computational Analysis

4.1 Simulation design

We conducted a simulation study to evaluate the performance of Balanced Randomization (BR) relative to Monte Carlo permutation (MC) and nonparametric bootstrap (BA) procedures in balanced two-sample testing problems. Throughout, MC and BA were implemented with $S = 10^4$ resamples (denoted $S_{\text{MC}} = S_{\text{BA}} = 10^4$ in the theoretical notation), whereas BR was implemented using fixed exact or near-balanced supports constructed in advance for each (N, n) configuration. The BR support size was approximately $|\mathcal{B}_R| \approx 300$, depending on the (N, n) configuration. This choice was made to balance computational efficiency against numerical resolution of the resulting BR p -values: $|\mathcal{B}_R| \approx 300$ yields a grid width of approximately $1/300 \approx 0.0033$, which is generally adequate for the nominal significance levels considered in this study, while still keeping the computational burden far below that of the stochastic resampling procedures used for comparison. The test statistic was the standard two-sample t -statistic.

For the main simulation study, we treat FP as a high-precision reference approximation to the full permutation test. For each simulated dataset, FP is computed using $S_{\text{FP}} = 10^6$ allocations drawn uniformly from $\mathcal{A}_{N,n}$. This large Monte Carlo sample is common across all regimes and provides a near-exact approximation to the full permutation distribution, while remaining computationally feasible even when $\binom{N}{n}$ is large. In smaller configurations, $\binom{N}{n}$ is sufficiently modest that exhaustive enumeration and the large- S Monte Carlo benchmark are numerically indistinguishable at the precision reported here, so we use the Monte Carlo implementation uniformly for simplicity.

Data were generated from five distributions chosen to represent both symmetric and skewed settings: standard Normal, Uniform[0, 1], Exponential with rate 1, Gamma with shape 2 and scale 1, and Lognormal with parameters (0, 1). We considered three total sample-size regimes: small ($11 \leq N \leq 20$), moderate ($21 \leq N \leq 30$), and large ($31 \leq N \leq 40$). Group sizes were taken to be as balanced as possible, with $n_1 = n_2 = N/2$ when N was even and the nearest balanced split otherwise.

For each combination of distribution, sample size N , and hypothesis setting, we generated $M = 200$ independent datasets. Under H_0 , the two groups were sampled from the same distribution. Under H_1 , a location shift corresponding to Cohen's $d = 0.5$ was introduced by adding 0.5σ to one

group, where σ denotes the population standard deviation of the generating distribution. For non-normal models, σ was computed from the corresponding model parameters. To reflect simulation variability, the stochastic comparators MC and BA were regenerated independently across replicates, whereas BR was evaluated on a fixed support for each (N, n) configuration.

All simulations were implemented in R using custom code for BR construction together with standard implementations for permutation and bootstrap procedures. For the Balanced Randomization construction, near-balanced supports were generated by a greedy optimization heuristic based on the discrepancy criterion in Step 3 of Section 2.1, with weights $w_1 = 1$ and $w_2 = 1$. This equal weighting was adopted as a simple default that simultaneously controls marginal and pairwise balance across settings; alternative weighting schemes that place greater emphasis on second-order balance are a natural topic for future investigation. For reproducibility, simulation scripts set explicit random seeds at the start of each experiment; the corresponding seed values and scripts are provided in the accompanying code repository.

To assess robustness beyond the t -statistic setting, we also repeated the simulation study using the Wilcoxon rank-sum statistic. The simulation design was identical to that described above, including the same data-generating distributions, sample-size regimes, and hypothesis configurations. For each (N, n) configuration we used a fixed, interpretable BR support in the Wilcoxon study, chosen to parallel the t -statistic designs while allowing direct comparison of approximation behavior across statistics without changing the underlying allocation structure.

4.2 Evaluation metrics

To assess approximation quality relative to the exact permutation benchmark, we used two primary metrics.

First, for stochastic resampling methods, we computed the nominal containment proportion

$$CP = \frac{1}{M} \sum_{i=1}^M \mathbf{1}\{p_i \in [L_i, U_i]\},$$

where

$$L_i = \hat{p}_i - z_{0.975} \sqrt{\frac{\hat{p}_i(1 - \hat{p}_i)}{S}}, \quad U_i = \hat{p}_i + z_{0.975} \sqrt{\frac{\hat{p}_i(1 - \hat{p}_i)}{S}},$$

$\hat{p}_i$ is the approximate p -value, p_i is the exact benchmark p -value, S is the number of resamples, and $z_{0.975} \approx 1.96$. For MC and BA, values of CP near 0.95 indicate approximation quality consistent with nominal Monte Carlo variability. This follows from the approximate binomial variance of the Monte Carlo estimator $\hat{p}_i$ with S resamples, which motivates the normal-approximation intervals used in the definition of CP .

Second, as a pointwise accuracy measure, we computed the root mean square error

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_{i=1}^M (\hat{p}_i - p_i)^2}.$$

Lower RMSE indicates more accurate approximation to the exact permutation benchmark.

Interpretation for BR. Because the BR support $\mathcal{B}_R$ is deterministic for fixed $(N, n, \mathcal{D})$, the BR p -value has no resampling-induced variability:

$$\text{Var}(p_{\text{BR}} \mid X, \mathcal{D}) = 0.$$

Accordingly, interval-based containment summaries are not interpreted as confidence statements for BR itself. They are reported only as reference benchmarks for comparing deterministic BR approximation error with the simulation variability inherent in MC and BA. Our primary evaluation of BR therefore emphasizes RMSE and agreement with the exact permutation benchmark. In particular, we do not interpret CP as a nominal coverage diagnostic for BR, and we do not use it as a primary metric for comparing BR with stochastic methods. For BR, our main accuracy summary is RMSE relative to the permutation benchmark.

4.3 Results

Before evaluating empirical power, we assessed the calibration of the proposed method under the null hypothesis. Table 1 reports the empirical Type I error rates (rejection proportions at the nominal level $\alpha = 0.05$) across the five data-generating distributions. The empirical sizes for Balanced Randomization remain well-calibrated and consistent with both the nominal level and the Monte Carlo permutation benchmarks.

Table 1: Empirical Type I error rates at nominal level $\alpha = 0.05$ under the null hypothesis. Values are averaged over the simulation regimes considered in Section 4.

Distribution	BR	MC ($S = 10^4$)	BA ($S = 10^4$)
Normal	0.051	0.049	0.052
Uniform	0.048	0.050	0.047
Exponential	0.052	0.048	0.054
Gamma	0.049	0.051	0.053
Lognormal	0.050	0.049	0.055

Note. The values reported are averages over the small, moderate, and large sample-size regimes.

4.3.1 Power analysis

To complement p -value accuracy metrics, we evaluated empirical power, defined as the proportion of rejections at nominal level $\alpha = 0.05$ under H_1 . Table 2 reports power by distribution and sample-size regime.

Power increases with N across all methods. In the simulation settings considered here, BR frequently attains slightly higher rejection rates than MC and BA, with the largest differences appearing under skewed distributions. From a randomization-inference perspective, the full permutation test is the natural benchmark and defines the target power curve. The slightly higher empirical power observed for BR relative to MC and BA at a fixed resampling budget is therefore best interpreted as closer numerical agreement with this benchmark, not as power exceeding that of the exact test itself. These results should be interpreted together with the accuracy and agreement summaries below and with the design diagnostics reported in Appendix C.

4.3.2 Estimation accuracy

Table 3 summarizes RMSE relative to the exact permutation benchmark, averaged across sample-size regimes and hypothesis settings.

In the simulation settings considered here, BR attains the smallest RMSE across all five distributions, with the largest margins appearing under skewed distributions. MC performs competitively

Table 2: Empirical power at level 0.05 under H_1 with Cohen’s $d = 0.5$. Averages within regimes: Small ($11 \leq N \leq 20$, $\bar{N} \approx 16$); Moderate ($21 \leq N \leq 30$, $\bar{N} \approx 26$); Large ($31 \leq N \leq 40$, $\bar{N} \approx 36$). Higher values indicate greater empirical sensitivity in the simulation settings considered here.

Distribution	Regime	BR Support	MC ($S = 10^4$)	BA ($S = 10^4$)
Normal	Small	0.52	0.51	0.50
	Moderate	0.72	0.71	0.69
	Large	0.85	0.84	0.82
Uniform	Small	0.54	0.53	0.51
	Moderate	0.74	0.73	0.71
	Large	0.87	0.86	0.84
Exponential	Small	0.45	0.42	0.38
	Moderate	0.65	0.62	0.57
	Large	0.78	0.75	0.70
Gamma	Small	0.47	0.44	0.40
	Moderate	0.67	0.64	0.59
	Large	0.80	0.77	0.72
Lognormal	Small	0.44	0.41	0.36
	Moderate	0.64	0.61	0.55
	Large	0.77	0.74	0.68

Note. In the t -statistic experiments, BR used fixed exact supports for selected small (N, n) configurations and near-balanced supports with size approximately $|\mathcal{B}_R| \approx 300$ otherwise.

Table 3: RMSE of approximate p -values relative to exact permutation benchmarks, averaged across sample-size regimes and simulation settings. Lower values indicate more accurate approximation.

Distribution	BR ($ \mathcal{B}_R \approx 300$)	MC ($S = 10^4$)	BA ($S = 10^4$)
Normal	0.012	0.018	0.031
Uniform	0.011	0.017	0.029
Exponential	0.014	0.024	0.046
Gamma	0.015	0.026	0.049
Lognormal	0.016	0.028	0.053
Overall	0.014	0.023	0.042

BR used fixed exact or near-balanced supports with size approximately $|\mathcal{B}_R| \approx 300$.

but remains less accurate than BR at the fixed resampling budget used here, while BA shows larger approximation error, particularly in the asymmetric settings. These findings are consistent with the view that structured deterministic supports can improve approximation quality when the number of statistic evaluations is limited.

We also observed empirically that lower-discrepancy supports tended to yield more accurate p -value approximations, suggesting a practical connection between the design criterion $D(\mathcal{D})$ and downstream inferential accuracy. For linear allocation statistics, this empirical pattern is consistent with Proposition 2, which bounds finite-sample variance distortion in terms of pairwise inclusion imbalance and hence in terms of low-order discrepancy.

These results should not be interpreted as implying that a 300-point deterministic support universally dominates 10^4 random permutations. Rather, the reported gains arise from the particular low-discrepancy, near-balanced supports used in the present simulation design and should be viewed as empirical evidence for these configurations. In other settings or with different statistics, a substantially larger BR support or a larger Monte Carlo budget may be needed to achieve comparable approximation quality.

4.3.3 Rank-based results

To assess robustness beyond moment-based statistics, we also conducted a parallel approximation study using the Wilcoxon rank-sum statistic. The data-generating distributions, sample-size regimes, and hypothesis settings matched those used in the t -statistic simulations, while the resampling budget and support choices were adapted to the secondary robustness-study setting. The smaller Monte Carlo budget in the Wilcoxon experiment was chosen because the rank-based study is intended as a secondary robustness analysis rather than as the primary computational comparison. In contrast to the t -statistic experiment, we used fixed, interpretable BR supports for Wilcoxon: an exact BIBD-based support for $(16, 8)$, a 50-block combinatorial design for $(26, 13)$, and a near-balanced support of size approximately 300 for $(36, 18)$. These supports were held fixed across all Wilcoxon replicates, so that differences in approximation behavior reflect the choice of statistic rather than changes in the underlying allocation structure.

For the Wilcoxon rank-sum robustness study, the permutation benchmark is defined in a hybrid fashion. When the number of distinct allocations $\binom{N}{n}$ does not exceed a pre-specified threshold M_{exact} , we compute the exact permutation p -value by enumerating all allocations. When $\binom{N}{n} > M_{\text{exact}}$, we instead use a high-precision Monte Carlo benchmark based on $S_{\text{bench}} = 50,000$ uniformly sampled allocations. We denote this hybrid benchmark by “EX” in the Wilcoxon tables and treat it as the ground-truth reference for BR and MC comparisons. Thus EX is exact whenever full enumeration is feasible and otherwise serves as a high-precision approximation to the full permutation distribution.

In this rank-based setting, Monte Carlo permutation with $S_{\text{MC}} = 5,000$ resamples provided the smallest RMSE relative to the permutation benchmark across the configurations considered. The BR-based Wilcoxon p -values remained reasonably close to the permutation benchmark and exhibited empirical power comparable to that of MC, although their RMSE was consistently larger than that of MC at the fixed computational budget. By contrast, the bootstrap approximation was considerably less accurate for the Wilcoxon statistic in these settings, with RMSE values an order of magnitude larger than those of BR and MC across all generating distributions. These findings indicate that BR remains viable for a canonical rank-based statistic, while also showing that its relative advantage over stochastic Monte Carlo can depend on the statistic and on how strongly the BR support is optimized for the corresponding reference distribution. Detailed Wilcoxon summaries, including RMSE and power by distribution and sample-size regime, are reported in Appendix C.

4.3.4 Hypothesis-testing agreement

To assess agreement with the exact permutation benchmark at the decision level, we modeled the binary indicator of correct reject/fail-to-reject agreement at level 0.05 using generalized estimating equations with exchangeable working correlation. Table 4 reports risk ratios comparing BR against MC and BA.

Table 4: Risk ratios for agreement with the exact permutation benchmark at level 0.05. Values are estimated from a GEE model; larger values indicate higher agreement. Comparator agreement gives the observed agreement rate for the comparison method.

Comparison	RR	95% CI	Comparator agreement
BR vs. MC	1.03	(1.01, 1.05)	0.931
BR vs. BA	1.20	(1.15, 1.26)	0.882

Monte Carlo uncertainty in the reported summaries was modest at the chosen replication size $M = 200$; for the principal proportion-based summaries, the corresponding standard errors were small enough that minor simulation noise does not materially affect the main qualitative comparisons. The GEE-based risk ratios in Table 4 should therefore be viewed as convenient summary statistics for agreement behavior under the present simulation design, rather than as stand-alone inferential targets.

The estimated risk ratio relative to bootstrap indicates that, in the simulation scenarios considered here, BR more frequently agrees with the exact permutation decision at the nominal 0.05 level. The corresponding improvement relative to MC is smaller but still positive. These agreement-based summaries should be interpreted as empirical performance measures tied to the present simulation design rather than as stand-alone measures of theoretical validity.

Where reported, standardized effect-size summaries for agreement proportions were moderate for BR versus BA and small for BR versus MC, which is consistent with the relative magnitudes of the estimated risk ratios. Additional diagnostics and distribution-specific summaries are provided in Appendix C.

4.4 Computational analysis

For a fixed BR support of size $|\mathcal{B}_R|$, the dominant computational cost is the evaluation of the test statistic over approximately $|\mathcal{B}_R|$ allocations, compared with 10^4 resamples for MC and BA in the present study. Thus, when $|\mathcal{B}_R| \ll 10^4$, BR substantially reduces the number of statistic evaluations required for inference.

Empirical timings in Table 12 reflect this reduction in computational burden. In the simulation settings considered here and in the computing environment used, BR yields markedly lower runtime than MC and BA while remaining close to the exact permutation benchmark in both RMSE and decision agreement. This trade-off may be especially useful in applications involving repeated testing or large collections of features, provided that the design-construction stage remains computationally modest relative to statistic evaluation.

In all reported timings, the one-time cost of BR design construction is excluded, since the same support can be reused across many datasets or features for a given (N, n) configuration. In the simulation settings considered here, this design-construction stage was modest relative to repeated statistic evaluations, especially in high-dimensional settings where the same BR support is applied feature-wise.

5 Comparison with Modern Resampling Methods

To place Balanced Randomization (BR) in a broader methodological context, we briefly compare it with several modern resampling approaches used in nonparametric and semiparametric inference: the wild bootstrap, subsampling, and variance-reduced random-permutation methods. The purpose of this section is not to claim universal superiority, but rather to clarify the inferential setting for which BR is designed and the ways in which it differs from existing approaches.

5.1 Wild bootstrap

The wild bootstrap is widely used in regression problems with heteroscedastic errors, where resampling is performed through transformed residuals rather than by reallocating observations directly (22; 12). In a generic regression setting, bootstrap responses take the form

$$Y_i^* = \hat{Y}_i + \hat{\varepsilon}_i w_i,$$

where the weights w_i are independent random variables satisfying

$$\mathbb{E}(w_i) = 0, \quad \text{Var}(w_i) = 1.$$

BR is not intended to replace the wild bootstrap in regression or heteroscedastic modeling settings. Rather, the two methods address different inferential targets. The wild bootstrap is justified through asymptotic arguments tied to a fitted regression structure, whereas BR is designed to approximate the finite-sample permutation distribution induced by exchangeability of the observed labels. In that sense, BR is most naturally viewed as a model-free alternative for permutation-style inference rather than as a substitute for regression-based bootstrap procedures. Formally, BR targets the permutation distribution of the test statistic under exchangeability of the observed labels, whereas wild bootstrap procedures target asymptotic sampling distributions under parametric or semiparametric regression assumptions.

5.2 Subsampling

Subsampling approximates the distribution of a statistic by recomputing it over subsamples of size $m < n$ drawn without replacement (16). Let

$$T_m^{(b)}, \quad b = 1, \dots, B,$$

denote the statistic computed from the b -th subsample. The empirical distribution of these values is then used to approximate the sampling distribution of the full-sample statistic.

Subsampling can be valid under broad conditions, but its practical performance depends on the choice of the tuning parameter m , and finite-sample behavior may be sensitive to that choice. Relative to subsampling, BR retains full-sample allocations and instead approximates the full permutation reference distribution through exact or near-balanced supports. The two methods therefore differ both in construction and in inferential target: subsampling is a reduced-sample approximation method, whereas BR remains within the full-sample permutation framework.

5.3 Random-permutation estimators

A number of methods have been proposed to reduce Monte Carlo variability in permutation testing, including antithetic sampling, stratified permutation sampling, and sequential or adaptive stopping

rules (1; 5; 7). These approaches remain stochastic, but they aim to improve efficiency relative to naive random sampling of permutations. A generic Monte Carlo permutation estimator takes the form

$$\hat{p}_S = \frac{1}{S} \sum_{s=1}^S \mathbf{1}\{|T_s| \geq |T_{\text{obs}}|\},$$

where the approximation error depends in part on the randomness of the sampled permutations.

BR is closest in spirit to this class of methods. The main distinction is that, whereas stratified or antithetic permutation schemes seek to reduce Monte Carlo variability through randomized design features, BR instead uses exact or near-balanced combinatorial supports to approximate the permutation space deterministically through low-order inclusion balance. Once the support is fixed, BR yields a reproducible reference distribution and a reproducible p -value.

5.4 Positioning of BR

The practical role of BR is therefore narrower but also more specific than that of a generic resampling method. It is designed for settings in which:

- permutation-style inference under exchangeability is the inferential target;
- exhaustive enumeration is computationally impractical;
- one seeks to reduce simulation variability without leaving the permutation framework.

Table 5 summarizes these distinctions at a qualitative level.

Table 5: Qualitative comparison of BR with several modern resampling approaches. The entries are intended as broad methodological descriptors rather than universal rankings across all applications.

Method	Primary setting	Full-sample	Determ. support	Typical role
Wild bootstrap	Regression; heteroscedasticity error	Yes	No	Model-based inference
Subsampling	General asymptotic approx.	No	No	General-purpose consistency
Random-perm. estimators	Permutation testing	Yes	No	Efficient stochastic test
MC permutation	Permutation testing	Yes	No	Baseline stochastic test
BR	Exchangeable permutations	Yes	Yes	Deterministic permutation approximation

Note. The entries in Table 5 are intended as broad descriptors of inferential target and support construction, rather than as rankings across all implementations or applications.

In this sense, BR should be viewed as a specialized alternative within the broader family of permutation-based approximation methods, rather than as a replacement for all modern resampling procedures.

6 High-Dimensional Illustration

To illustrate the practical behavior of Balanced Randomization (BR) in a large-scale testing problem, we considered a simulated genomic-style dataset with $N = 40$ observations split into two balanced

groups of size 20, and $p = 5000$ features. Among these features, 10 contained true location shifts with effect sizes $d \in \{0.5, 1\}$, while the remaining features were null. A two-sample t -statistic was computed feature-wise, and the features were generated independently across $j = 1, \dots, p$ in this illustration.

In the high-dimensional illustration ($N = 40$, $n = 20$, $p = 5000$), we compare BR, a nonparametric bootstrap, and a permutation-based benchmark at the feature level. For each gene, we compute three p -values: (i) a BR value based on a near-balanced support with $b' = 300$ allocations, (ii) a bootstrap value (BA) based on $S_{\text{BA}} = 10^4$ group-wise resamples, and (iii) a high-precision permutation benchmark based on $S_{\text{MC,bench}} = 10^6$ allocations drawn uniformly from $\mathcal{A}_{N,n}$. This benchmark serves as the “permutation-based” reference in the agreement and runtime summaries, providing a close approximation to the full permutation distribution while avoiding exhaustive enumeration over $\binom{40}{20}$ allocations.

BR used exact or near-balanced supports of size approximately $|\mathcal{B}_R| \approx 300$, constructed by cyclic initialization with greedy refinement. For each simulation replicate, the same BR support for the given (N, n) configuration was applied feature-wise across all p variables, so that differences in feature-level p -values reflect variation in the data rather than changes in the underlying allocation structure. Agreement was evaluated both at the raw significance level $\alpha = 0.05$ and after Benjamini–Hochberg false discovery rate control at $q = 0.05$. Here, agreement denotes the proportion of feature-level reject/non-reject decisions matching those obtained from the high-precision permutation benchmark.

Table 6: Illustrative agreement and efficiency summaries for the high-dimensional illustration ($p = 5000$, $N = 40$). Agreement rates are reported relative to a high-precision permutation benchmark. Relative runtime is normalized to the BA procedure (100%).

Method	Raw agreement	FDR agreement	Relative runtime
BR ($ \mathcal{B}_R \approx 300$)	94–97%	92–96%	3–5%
BA ($S = 10^4$)	82–88%	80–85%	100%
Benchmark ($S = 10^6$)	100%	100%	–

The ranges in Table 6 are taken over several independent simulation replicates of the high-dimensional setting and are intended to indicate typical variability across runs, rather than single-instance values. Additional supplementary summaries, including Jaccard indices for the raw and Benjamini–Hochberg adjusted rejection sets, are reported in Appendix C; see Table 13.

The high-precision permutation benchmark required a substantially larger computational budget than either BA or BR and is included primarily as a reference standard for accuracy rather than as a practical competitor in large-scale screening.

In this illustrative setting, BR showed high agreement with the high-precision permutation benchmark while requiring substantially fewer statistic evaluations than the stochastic alternatives considered. The gains were evident in these simulations both for raw feature-level testing and after false discovery rate adjustment, indicating that the approximation remained stable under multiplicity correction in this design.

Because the BR support is fixed once constructed, the resulting feature-level p -values, rankings, and rejection sets are reproducible across repeated runs of the same analysis. This is especially useful for threshold-based procedures such as Benjamini–Hochberg adjustment, where small stochastic perturbations in borderline p -values can alter the final discovery set. In contrast, stochastic resampling methods may produce slight run-to-run variation unless very large resampling budgets are

used.

This example is intended as a practical illustration rather than as a separate multiple-testing theory for BR. Its role is to show that deterministic structured supports can remain useful in large-scale nonparametric screening problems where exhaustive permutation evaluation is infeasible but close agreement with permutation-style inference is still desired.

7 Discussion

The Balanced Randomization (BR) framework developed in this paper provides a structured, design-based approximation to permutation inference for settings in which exhaustive enumeration is computationally impractical. Its central features are deterministic reproducibility, low-order inclusion balance, and a direct connection to the permutation reference distribution under exchangeability. In the theoretical development, these properties are formalized through the well-posedness of the BR support, the absence of resampling-induced variability (Lemma 1), moment matching for linear allocation statistics (Proposition 1), and a sufficient condition for asymptotic validity for asymptotically linear statistics under suitable regularity conditions (Theorem 1).

The numerical results suggest that these structural properties can translate into favorable empirical performance under fixed computational budgets. In the settings studied here, BR achieved smaller RMSE than the stochastic alternatives considered, with the largest differences appearing under skewed distributions, and it also showed higher agreement with the exact or high-precision permutation benchmark at the testing-decision level. These gains were obtained while using substantially fewer statistic evaluations than the Monte Carlo and bootstrap comparators considered in the study. Taken together, these findings suggest that structured deterministic sampling can improve approximation quality at a fixed computational budget, rather than relying on the larger resampling budgets typically needed to reduce stochastic simulation error.

At a conceptual level, this deterministic-support perspective is also related in spirit to quasi-Monte Carlo methodology, where carefully constructed low-discrepancy point sets are used in place of random sampling to improve numerical approximation of expectations (13). The analogy is heuristic rather than formal, since BR operates on combinatorial allocation spaces and uses inclusion-probability discrepancy rather than geometric discrepancy to construct its supports. Nevertheless, both approaches reflect the same principle: deterministic structure can improve approximation stability for a fixed computational budget. The Wilcoxon rank-sum study in Section 4 provides a complementary robustness check, showing that BR can be applied to a canonical rank-based statistic, while in that setting Monte Carlo permutation achieved smaller RMSE at the chosen resampling budget. This reinforces the view that the relative advantage of BR versus Monte Carlo can depend on the choice of test statistic and on how strongly the BR supports are optimized for the corresponding reference distribution. This is consistent with the scope of Theorem 1, which is formulated for asymptotically linear statistics and does not claim universal asymptotic optimality for arbitrary nonlinear functionals.

The practical appeal of BR is especially evident in settings where reproducibility matters. Conditional on the chosen BR support, repeated analyses under the same configuration yield the same reference distribution and the same p -values. In large-scale testing problems, this stability extends to feature rankings and threshold-based discovery sets, which can be sensitive to small stochastic perturbations in approximate p -values when resampling budgets are limited. Such reproducibility is valuable in collaborative and regulatory settings, where the ability to replicate statistical results exactly across repeated analyses can be important.

At the same time, the scope of the present work should be stated clearly. The role of BR

is therefore specialized: it serves as a deterministic approximation method for permutation-style inference under exchangeability, particularly in settings where exhaustive permutation testing is infeasible but close agreement with the permutation benchmark is still desired. BR is not proposed as a replacement for all modern resampling methods, nor does the current theory imply universal finite-sample superiority over stochastic alternatives.

Several limitations should be noted. First, exact BIBD constructions are available only for restricted parameter combinations, so practical implementation often relies on near-balanced supports obtained by heuristic optimization. Second, the strongest finite-sample moment-matching statements in this paper are developed for linear allocation statistics, and additional work is needed to characterize approximation quality more broadly for nonlinear statistics. Third, although the high-dimensional illustration is encouraging, it is intended as empirical evidence rather than a general multiple-testing theory for BR-based procedures. BR is not intended for settings in which exchangeability fails, such as time-series, clustered, or otherwise dependent data without appropriate adjustment, nor for statistics whose behavior depends strongly on higher-order dependence features not well captured by first- and second-order inclusion balance. The Wilcoxon results further indicate that approximation quality may vary across statistics, especially when the BR support is not optimized for the specific statistic being evaluated.

Several extensions merit further study. These include constructions for multi-sample and more general allocation problems, covariate-adjusted or regression-based permutation procedures, adaptive selection of support size based on discrepancy diagnostics, and structured designs for dependent or constrained data settings. Each of these directions would broaden the applicability of the BR framework while preserving its central design-based motivation.

Overall, the results indicate that structured deterministic supports can provide a practically useful middle ground between exhaustive permutation inference and stochastic resampling in the exchangeable settings studied here.

8 Conclusion

Balanced Randomization offers a deterministic, structured approximation to permutation inference when exhaustive evaluation of the allocation space is computationally impractical. In the settings studied here, it achieved close agreement with exact or high-precision permutation benchmarks while requiring substantially fewer statistic evaluations than the stochastic alternatives considered. By leveraging low-order moment matching through exact or near-balanced combinatorial supports, the method approximates the inclusion structure of the full allocation space with substantially reduced computational effort.

The main contribution of the BR framework is not exactness in general, but reproducible approximation through carefully constructed exact or near-balanced supports, whose finite-sample performance can depend on the statistic and design used. This makes BR particularly attractive in applications where simulation variability, computational cost, and run-to-run stability are all important. In large-scale simultaneous inference settings, the deterministic nature of BR can provide the stability needed for reproducible feature discovery across repeated analyses and research pipelines. Future work will focus on extending the framework to multi-sample problems, regression-type permutation procedures, and design constructions tailored to more complex dependence structures. An important complementary direction is to develop sharper theoretical guarantees for near-balanced supports and to characterize more fully how BR performance depends on the choice of test statistic beyond the asymptotically linear case.

Acknowledgments

The authors thank Tulane University and USTTB for institutional support.

Code and data availability

All data used in this study are simulated. The complete simulation code, data generation scripts, analysis pipelines, and supplementary results required to reproduce the tables and figures in this paper are publicly available at <https://github.com/sudeshsrivastav/BR-Resampling>. The repository includes scripts for BR design construction, Monte Carlo permutation and bootstrap comparisons, simulation summaries, and documentation for reproducing the reported results.

Appendix A. Proofs and proof sketches

A.1 Well-posedness of the BR support

Definition 1 introduces the Balanced Randomization (BR) support

$$\mathcal{B}_R = \text{Unique}\{g(B_i) : g \in \mathcal{G}, i = 1, \dots, b\} \subseteq \mathcal{A}_{N,n},$$

where

$$\mathcal{A}_{N,n} = \{A \subseteq \{1, \dots, N\} : |A| = n\}.$$

We verify that this construction is well defined.

First, for each $i \in \{1, \dots, b\}$, the block B_i is an element of $\mathcal{A}_{N,n}$ and hence has cardinality n . Since each $g \in \mathcal{G}$ acts as a permutation of $\{1, \dots, N\}$, it preserves set cardinality. Therefore $g(B_i) \subseteq \{1, \dots, N\}$ and $|g(B_i)| = |B_i| = n$, so $g(B_i) \in \mathcal{A}_{N,n}$.

Second, the collection

$$\{g(B_i) : g \in \mathcal{G}, i = 1, \dots, b\}$$

is finite because both $\mathcal{G}$ and $\{1, \dots, b\}$ are finite. Removing duplicates therefore yields a finite set $\mathcal{B}_R \subseteq \mathcal{A}_{N,n}$, so the support cardinality $|\mathcal{B}_R|$ is well defined.

Third, because $\mathcal{B}_R$ is finite and nonempty, the empirical distribution

$$\hat{F}_{BR}(t) = \frac{1}{|\mathcal{B}_R|} \sum_{A \in \mathcal{B}_R} \mathbf{1}\{T(A) \leq t\}$$

is a valid distribution function on the image of T over $\mathcal{B}_R$. Likewise, for a two-sided test, the BR p -value

$$p_{BR} = \frac{1}{|\mathcal{B}_R|} \sum_{A \in \mathcal{B}_R} \mathbf{1}\{|T(A)| \geq |T_{\text{obs}}|\}$$

is well defined and belongs to the interval $[0, 1]$.

Thus Definition 1 is mathematically well posed. □

A.2 Proof of Lemma 1

Fix observed data X , a design $\mathcal{D}$, and a group action $\mathcal{G}$. By Definition 1, the BR support

$$\mathcal{B}_R = \text{Unique}\{g(B_i) : g \in \mathcal{G}, i = 1, \dots, b\}$$

is completely determined by $(\mathcal{D}, \mathcal{G})$. Once X is also fixed, the observed statistic T_{obs} and every allocation-specific value $T(A)$, for $A \in \mathcal{B}_R$, are deterministic.

Therefore each indicator

$$\mathbf{1}\{|T(A)| \geq |T_{\text{obs}}|\}, \quad A \in \mathcal{B}_R,$$

is deterministic conditional on $(X, \mathcal{D}, \mathcal{G})$, and hence so is their average,

$$p_{\text{BR}} = \frac{1}{|\mathcal{B}_R|} \sum_{A \in \mathcal{B}_R} \mathbf{1}\{|T(A)| \geq |T_{\text{obs}}|\}.$$

A deterministic quantity has conditional variance zero, so

$$\text{Var}(p_{\text{BR}} \mid X, \mathcal{D}, \mathcal{G}) = 0.$$

This proves Lemma 1. □

A.3 Proof of Proposition 1

Recall that

$$T(A) = \sum_{i=1}^N a_i Z_i(A), \quad Z_i(A) = \mathbf{1}\{i \in A\}, \quad \sum_{i=1}^N a_i = 0.$$

We first compare expectations. Under any probability measure on $\mathcal{A}_{N,n}$,

$$E\{T(A)\} = E\left[\sum_{i=1}^N a_i Z_i(A)\right] = \sum_{i=1}^N a_i E\{Z_i(A)\}.$$

Hence the mean depends only on the first-order inclusion probabilities $P(i \in A)$. By assumption,

$$P_{\text{BR}}(i \in A) = \frac{n}{N}$$

for all i , and these are exactly the first-order inclusion probabilities under the uniform distribution on $\mathcal{A}_{N,n}$. Therefore

$$E_{\text{BR}}\{T(A)\} = E\{T(A)\}.$$

For the variance, write

$$\text{Var}\{T(A)\} = \text{Var}\left(\sum_{i=1}^N a_i Z_i(A)\right) = \sum_{i=1}^N a_i^2 \text{Var}\{Z_i(A)\} + 2 \sum_{1 \leq i < j \leq N} a_i a_j \text{Cov}\{Z_i(A), Z_j(A)\}.$$

Now

$$\text{Var}\{Z_i(A)\} = P(i \in A)(1 - P(i \in A)),$$

and for $i \neq j$,

$$\text{Cov}\{Z_i(A), Z_j(A)\} = P(i, j \in A) - P(i \in A)P(j \in A).$$

Thus the variance depends only on the first- and second-order inclusion probabilities. By assumption,

$$P_{\text{BR}}(i \in A) = \frac{n}{N}, \quad P_{\text{BR}}(i, j \in A) = \frac{n(n-1)}{N(N-1)},$$

which match the corresponding probabilities under the uniform distribution on $\mathcal{A}_{N,n}$. Therefore the variances also coincide:

$$\text{Var}_{\text{BR}}\{T(A)\} = \text{Var}\{T(A)\}.$$

This proves Proposition 1. □

A.4 Proof sketch for Theorem 1

We sketch the argument in the notation of Theorem 1. Write

$$T_N(A) = L_N(A) + R_N(A), \quad L_N(A) = \sum_{i=1}^N a_{Ni} Z_i(A), \quad \sum_{i=1}^N a_{Ni} = 0.$$

The proof proceeds in three steps.

Step 1: Low-order inclusion matching for the linear part. By assumption,

$$\max_i \left| P_{\text{BR},N}(i \in A) - \frac{n}{N} \right| \rightarrow 0$$

and

$$\max_{i \neq j} \left| P_{\text{BR},N}(i, j \in A) - \frac{n(n-1)}{N(N-1)} \right| \rightarrow 0.$$

Because the mean and variance of the linear statistic $L_N(A)$ depend only on first- and second-order inclusion probabilities, these conditions imply that the BR mean and variance of L_N are asymptotically equal to those under the uniform distribution on $\mathcal{A}_{N,n}$. In particular,

$$E_{\text{BR},N}\{L_N(A)\} - E\{L_N(A)\} \rightarrow 0$$

and

$$\text{Var}_{\text{BR},N}\{L_N(A)\} - \text{Var}\{L_N(A)\} \rightarrow 0,$$

under the scaling regime of the theorem.

Step 2: Asymptotic agreement for the standardized linear part. Under the theorem's assumptions, the standardized version of the linear part satisfies a combinatorial central limit theorem under the full allocation distribution. Since the BR support reproduces the relevant first- and second-order inclusion structure asymptotically, the corresponding BR-standardized linear statistic has the same asymptotic centering and scaling. Consequently, the BR and full-allocation versions of the standardized linear part share the same limiting normal distribution.

At an intuitive level, the BR support is sufficiently balanced that, for asymptotically linear statistics, it preserves the low-order dependence structure governing the Gaussian limit. This yields asymptotic equivalence of critical values and tail probabilities for the linear part.

Step 3: Transfer from L_N to T_N . By assumption, the remainder term $R_N(A)$ is negligible relative to the standard deviation of $L_N(A)$, uniformly over allocations in the relevant support. Hence

$$\frac{T_N(A) - L_N(A)}{\sqrt{\text{Var}\{L_N(A)\}}} \rightarrow 0$$

in the sense required for asymptotic replacement. A Slutsky-type argument then implies that T_N and L_N have asymptotically equivalent standardized distributions under both the BR and full-allocation measures.

Combining the three steps, the BR distribution of the standardized statistic is asymptotically equivalent to its full-allocation counterpart, and BR critical values and BR p -values are asymptotically valid for T_N .

Remark on scope. This proof sketch is intentionally limited to asymptotically linear statistics. A complete proof for broader nonlinear statistics would require additional arguments controlling higher-order dependence terms and the effect of imperfect balance beyond second order. Such extensions are beyond the scope of the present paper. $\square$

Appendix B. Construction heuristics for near-balanced supports

This appendix describes the heuristic procedures used to construct near-balanced supports when an exact $2-(N, n, \lambda)$ design is unavailable. Throughout, we work on the allocation space

$$\mathcal{A}_{N,n} = \{A \subseteq \{1, \dots, N\} : |A| = n\},$$

and seek a finite collection

$$\mathcal{D} = \{B_1, \dots, B_m\} \subseteq \mathcal{A}_{N,n}$$

whose first- and second-order inclusion behavior approximates that of the uniform distribution on $\mathcal{A}_{N,n}$.

B.1 Design objective

For a candidate support $\mathcal{D} \subseteq \mathcal{A}_{N,n}$, define the empirical first-order inclusion frequencies

$$\hat{\pi}_i(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{B \in \mathcal{D}} \mathbf{1}\{i \in B\}, \quad i = 1, \dots, N,$$

and the empirical second-order inclusion frequencies

$$\hat{\pi}_{ij}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{B \in \mathcal{D}} \mathbf{1}\{i \in B, j \in B\}, \quad 1 \leq i < j \leq N.$$

Under the uniform distribution on $\mathcal{A}_{N,n}$, the target inclusion probabilities are

$$\pi_i^* = \frac{n}{N}, \quad \pi_{ij}^* = \frac{n(n-1)}{N(N-1)}.$$

The ideal exact-balance conditions are therefore

$$\hat{\pi}_i(\mathcal{D}) = \pi_i^* \quad \text{for all } i, \quad \hat{\pi}_{ij}(\mathcal{D}) = \pi_{ij}^* \quad \text{for all } i < j.$$

When exact balance is unattainable, we instead minimize discrepancy from these targets.

A convenient objective is

$$D(\mathcal{D}) = w_1 \sum_{i=1}^N \left(\hat{\pi}_i(\mathcal{D}) - \frac{n}{N} \right)^2 + w_2 \sum_{1 \leq i < j \leq N} \left(\hat{\pi}_{ij}(\mathcal{D}) - \frac{n(n-1)}{N(N-1)} \right)^2,$$

where $w_1, w_2 \geq 0$ are tuning weights. Smaller values of $D(\mathcal{D})$ indicate that the selected support more closely reproduces the first- and second-order inclusion behavior of the full allocation space. In the implementations used in this paper, we set $w_1 = w_2 = 1$ as a simple default that balances marginal and pairwise inclusion control simultaneously. More generally, larger values of w_2 may be used when second-order balance is to be emphasized more strongly. Larger values of w_1 instead enforce more uniform marginal representation across indices.

The criterion $D(\mathcal{D})$ penalizes departures from the low-order inclusion structure of the full allocation distribution. Because the mean and variance of linear allocation statistics depend only on first- and second-order inclusion behavior, supports with smaller discrepancy are expected to yield better approximation for such statistics. Equivalently, a good near-balanced design has an inclusion matrix whose row sums are nearly constant and whose pairwise column co-occurrence frequencies are close to their target values under the uniform allocation distribution.

B.2 Greedy exchange heuristic

The optimization problem over subsets of $\mathcal{A}_{N,n}$ is combinatorial and generally infeasible to solve exactly for moderate N . We therefore use a greedy exchange heuristic with optional random restarts. This procedure is akin to a hill-climbing algorithm on the space of candidate supports, with optional random restarts used to mitigate sensitivity to local minima.

Initialization. Fix a target support size m . Construct an initial collection

$$\mathcal{D}^{(0)} = \{B_1^{(0)}, \dots, B_m^{(0)}\} \subseteq \mathcal{A}_{N,n}$$

using one of the following strategies:

1. random sampling without replacement from $\mathcal{A}_{N,n}$;
2. cyclic initialization from a seed block when N and n admit a natural shift structure;
3. seeding from a known exact design for nearby parameters, followed by augmentation or truncation.

Local update step. At iteration t , choose a candidate block $B \in \mathcal{D}^{(t)}$ and propose replacing it by a block $B' \in \mathcal{A}_{N,n} \setminus \mathcal{D}^{(t)}$. The updated support is

$$\mathcal{D}' = (\mathcal{D}^{(t)} \setminus \{B\}) \cup \{B'\}.$$

Accept the move if

$$D(\mathcal{D}') < D(\mathcal{D}^{(t)}).$$

If no improving move is found in a prescribed search neighborhood, the algorithm terminates or restarts from a new initialization.

Neighborhoods. To reduce computation, we do not search the entire space $\mathcal{A}_{N,n}$ at each iteration. Instead, we generate proposals B' from restricted neighborhoods, for example:

1. *single-swap proposals*, obtained by removing one element from B and replacing it by one element outside B ;
2. *double-swap proposals*, obtained by two such exchanges;
3. *random proposals*, sampled uniformly from $\mathcal{A}_{N,n}$.

Single-swap moves are often sufficient for local improvement, while occasional random proposals help escape poor local minima.

B.3 Incremental updating of the discrepancy

Direct recomputation of $D(\mathcal{D})$ after each proposed exchange can be expensive. To accelerate the search, we maintain running inclusion counts

$$c_i(\mathcal{D}) = \sum_{B \in \mathcal{D}} \mathbf{1}\{i \in B\}, \quad c_{ij}(\mathcal{D}) = \sum_{B \in \mathcal{D}} \mathbf{1}\{i \in B, j \in B\},$$

so that

$$\hat{\pi}_i(\mathcal{D}) = \frac{c_i(\mathcal{D})}{|\mathcal{D}|}, \quad \hat{\pi}_{ij}(\mathcal{D}) = \frac{c_{ij}(\mathcal{D})}{|\mathcal{D}|}.$$

If one block B is replaced by B' , then only the counts involving elements in $B \cup B'$ change. Thus the discrepancy difference

$$D(\mathcal{D}') - D(\mathcal{D})$$

can be computed from local count updates rather than recomputing all inclusion frequencies from scratch. This substantially reduces runtime in the design-construction stage.

B.4 Optional marginal-balance enforcement

In some implementations, we enforce approximate or exact marginal balance before optimizing second-order behavior. One simple strategy is to constrain the support so that the counts $c_i(\mathcal{D})$ differ by at most one across indices i . This guarantees

$$\max_{1 \leq i \leq N} \left| \hat{\pi}_i(\mathcal{D}) - \frac{n}{N} \right| \leq \frac{1}{|\mathcal{D}|},$$

up to the rounding implied by integer counts. Optimization can then focus primarily on second-order discrepancy.

B.5 Group-based expansion

After a base design $\mathcal{D}$ has been constructed, we may enlarge it using a group action $\mathcal{G}$ on $\{1, \dots, N\}$, as in Definition 1. The expanded support is

$$\mathcal{B}_R = \text{Unique}\{g(B) : g \in \mathcal{G}, B \in \mathcal{D}\}.$$

This step can improve coverage of the allocation space while preserving the structured nature of the construction. In practice, cyclic groups generated by modular shifts are particularly convenient when the index set admits a natural circular ordering.

When group expansion is used, discrepancy may be evaluated either on the base support $\mathcal{D}$ or on the expanded support $\mathcal{B}_R$, depending on whether the group action is regarded as part of the construction or as a post-processing step. In this paper, all reported BR p -values are computed on the final deduplicated support actually used for inference. Reported BR runtimes refer to evaluation over the final support; support-construction time was modest relative to repeated statistic evaluation and is documented in the reproducibility materials.

B.6 Practical stopping rules

The heuristic search is terminated when one of the following conditions is met:

1. a prescribed maximum number of iterations is reached;

2. no improving move is found after a fixed number of consecutive proposals;
3. the discrepancy $D(\mathcal{D})$ falls below a user-specified tolerance;
4. multiple random restarts fail to improve the best solution found so far.

In our numerical work, the chosen support sizes are deliberately modest, so the design-construction stage remains computationally manageable relative to the cost of evaluating test statistics across many resampling allocations.

B.7 Pseudocode

The following pseudocode summarizes the near-balanced construction.

1. **Input:** N , n , target support size m , tuning weights w_1, w_2 , iteration limits, and number of restarts.
2. **Initialize:** Construct an initial support $\mathcal{D}^{(0)} \subseteq \mathcal{A}_{N,n}$ of size m .
3. **Evaluate:** Compute $D(\mathcal{D}^{(0)})$ from first- and second-order inclusion frequencies.
4. **Iterate:**
 - (a) select a block $B \in \mathcal{D}^{(t)}$;
 - (b) generate one or more candidate replacements $B' \in \mathcal{A}_{N,n} \setminus \mathcal{D}^{(t)}$;
 - (c) accept an improving move if $D(\mathcal{D}') < D(\mathcal{D}^{(t)})$;
 - (d) otherwise retain $\mathcal{D}^{(t)}$ and continue.
5. **Restart if needed:** Repeat from several initial supports and retain the support with smallest discrepancy.
6. **Expand if desired:** Apply a group action and deduplicate to obtain the final support $\mathcal{B}_R$.
7. **Output:** Final support $\mathcal{B}_R$ together with its discrepancy summary.

In the numerical experiments reported in this paper, the greedy exchange heuristic was implemented with a maximum of 10,000 iterations per restart, 25 random restarts per (N, n) configuration, and at most 100 candidate proposals per iteration drawn from single- and double-swap neighborhoods. These values were chosen to keep design-construction time modest relative to repeated test-statistic evaluations while still producing supports with small low-order discrepancy.

B.8 Interpretation and limitations

The heuristic procedures in this appendix are intended to produce supports with small low-order discrepancy, not to solve the underlying combinatorial optimization problem exactly. Accordingly, they provide empirical design quality rather than a universal existence or optimality guarantee. Their role in the paper is practical: they supply structured supports that can be evaluated theoretically through low-order inclusion behavior and empirically through simulation performance.

For exact BIBD settings, these heuristics are unnecessary. Their main importance lies in extending the BR framework to parameter combinations for which no exact design is available or easily constructed.

Appendix C. Supplementary numerical results

This appendix collects supplementary numerical and diagnostic material intended to support, rather than define, the main methodology. In particular, the construction of the BR support, the standing assumptions for the theoretical results, and the principal simulation and application findings are given in the main text. The material reported here provides additional detail on low-order balance diagnostics, computational scaling, reference summaries for stochastic simulation error, and supplementary comparisons for the high-dimensional illustration.

Throughout, the exact benchmark is the uniform allocation distribution on

$$\mathcal{A}_{N,n} = \{A \subseteq \{1, \dots, N\} : |A| = n\},$$

whenever exhaustive enumeration is computationally feasible. In settings where exhaustive evaluation is impractical, the benchmark is a high-precision Monte Carlo approximation using a substantially larger number of resamples than the methods under comparison.

C.1 Design-quality diagnostics

To quantify the quality of a near-balanced support $\mathcal{D} \subseteq \mathcal{A}_{N,n}$, we report summaries of the empirical first- and second-order inclusion frequencies

$$\hat{\pi}_i(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{B \in \mathcal{D}} \mathbf{1}\{i \in B\}, \quad \hat{\pi}_{ij}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{B \in \mathcal{D}} \mathbf{1}\{i \in B, j \in B\},$$

relative to their target values

$$\pi_i^* = \frac{n}{N}, \quad \pi_{ij}^* = \frac{n(n-1)}{N(N-1)}.$$

For each constructed support, we summarize:

1. the maximum absolute first-order deviation

$$\Delta_1(\mathcal{D}) = \max_{1 \leq i \leq N} \left| \hat{\pi}_i(\mathcal{D}) - \frac{n}{N} \right|;$$

2. the maximum absolute second-order deviation

$$\Delta_2(\mathcal{D}) = \max_{1 \leq i < j \leq N} \left| \hat{\pi}_{ij}(\mathcal{D}) - \frac{n(n-1)}{N(N-1)} \right|;$$

3. the discrepancy criterion

$$D(\mathcal{D}) = w_1 \sum_{i=1}^N \left(\hat{\pi}_i(\mathcal{D}) - \frac{n}{N} \right)^2 + w_2 \sum_{1 \leq i < j \leq N} \left(\hat{\pi}_{ij}(\mathcal{D}) - \frac{n(n-1)}{N(N-1)} \right)^2.$$

Smaller values of $\Delta_1(\mathcal{D})$, $\Delta_2(\mathcal{D})$, and $D(\mathcal{D})$ indicate closer agreement with the low-order inclusion structure of the full allocation distribution. For linear allocation statistics, these quantities provide useful diagnostics for the quality of BR approximation.

These summaries are intended as diagnostics rather than formal guarantees. Their role is to document that the constructed supports are close, in low-order inclusion terms, to the target allocation distribution used by the exact permutation test.

Table 7: Supplementary design-quality summaries for near-balanced BR supports across sample-size regimes. Values are representative summaries over the supports used in the simulation study. Smaller values indicate closer agreement with the full allocation distribution.

Regime	$\Delta_1(\mathcal{D})$	$\Delta_2(\mathcal{D})$	$D(\mathcal{D})$
Small ($11 \leq N \leq 20$)	0.006	0.014	0.0018
Moderate ($21 \leq N \leq 30$)	0.004	0.010	0.0011
Large ($31 \leq N \leq 40$)	0.003	0.008	0.0007

Balanced Randomization supports for Wilcoxon rank-sum experiments

For the Wilcoxon rank-sum robustness study described in Section 4, we employed fixed, interpretable Balanced Randomization supports for each (N, n) configuration, rather than reoptimizing the designs specifically for the rank-based statistic. This subsection records the constructions used for $(N, n) = (16, 8)$ and $(26, 13)$.

For $(N, n) = (16, 8)$, the BR support was taken to be an exact 2-(16, 8, λ) balanced incomplete block design. In particular, we used a classical 2-(16, 8, 7) design with $v = 16$ elements, block size $k = 8$, and $b = 30$ blocks, so that each observation appears in exactly $r = 15$ blocks and each unordered pair of observations appears together in exactly $\lambda = 7$ blocks. The resulting design satisfies the standard BIBD relations

$$vr = bk, \quad \lambda(v - 1) = r(k - 1),$$

and induces exact first- and second-order inclusion probabilities matching those of the full allocation distribution for (16, 8).¹ In the Wilcoxon experiment, this BIBD was used as the BR support for all (16, 8) configurations, both under the null and under the alternative.

For $(N, n) = (26, 13)$, an exact BIBD with balanced pairwise concurrence is not available. Instead, we used a fixed near-balanced support consisting of $b = 50$ treatment allocations of size 13. This support was constructed offline by a discrete search heuristic designed to reduce the low-order discrepancy criterion in Section 2.1. The resulting design is not an exact BIBD, but it exhibits approximately balanced first- and second-order inclusion frequencies and serves as a structured, deterministic subset of the full allocation space. As with the (16, 8) case, this 54-block support was held fixed across all Wilcoxon replicates for (26, 13).

For $(N, n) = (36, 18)$, we used a near-balanced support of size approximately 300 constructed by the same discrepancy-based heuristic as in the t -statistic experiments. This design is not tailored specifically to Wilcoxon; it was reused to allow direct comparison of approximation behavior across statistics without changing the underlying resampling structure. In all cases, the Balanced Randomization Wilcoxon p -values were computed by evaluating the rank-sum statistic over the fixed BR support and taking the empirical two-sided tail probability with respect to this deterministic allocation distribution.

These choices reflect the role of the Wilcoxon experiment as a secondary robustness study: the supports are chosen to be interpretable and reasonably balanced, but they are not optimized specifically for the rank statistic.

Table 8 summarizes RMSE and empirical power for the Wilcoxon rank-sum experiments. In this setting, MC permutation yields the smallest RMSE across all configurations, whereas BR

¹The explicit block list is omitted here for brevity; it is available in the simulation code accompanying this manuscript.

Table 8: RMSE and empirical power at level 0.05 for the Wilcoxon rank-sum statistic under a location shift of Cohen’s $d = 0.5$. Results are shown by distribution and sample-size regime for Balanced Randomization (BR), Monte Carlo permutation (MC) with $S_{MC} = 5,000$ resamples, and the bootstrap approximation (BA). Lower RMSE indicates closer agreement with the permutation benchmark.

Distribution	Regime	RMSE _{BR}	RMSE _{MC}	RMSE _{BA}	Power _{BR}	Power _{MC}	Power _{BA}
Normal	Small	0.074	0.005	0.323	0.11	0.13	0.00
	Moderate	0.040	0.006	0.330	0.23	0.20	0.00
	Large	0.017	0.005	0.396	0.39	0.41	0.00
Uniform	Small	0.082	0.005	0.287	0.08	0.07	0.00
	Moderate	0.043	0.005	0.315	0.19	0.18	0.00
	Large	0.021	0.006	0.372	0.30	0.31	0.00
Exponential	Small	0.065	0.005	0.374	0.18	0.21	0.00
	Moderate	0.026	0.003	0.446	0.50	0.47	0.00
	Large	0.018	0.005	0.439	0.49	0.51	0.00
Gamma	Small	0.069	0.005	0.374	0.19	0.23	0.00
	Moderate	0.034	0.006	0.393	0.32	0.36	0.00
	Large	0.016	0.005	0.399	0.38	0.39	0.00
Lognormal	Small	0.045	0.004	0.494	0.38	0.44	0.00
	Moderate	0.023	0.003	0.503	0.70	0.72	0.00
	Large	0.009	0.002	0.513	0.88	0.88	0.00
Overall	—	0.054	0.015	0.692	0.34	0.36	0.00

Note. The empirical power of 0.00 for BA across all sub-regimes coincides with substantially inflated pointwise error ($\text{RMSE}_{BA} > 0.35$). This pronounced conservatism is qualitatively consistent with the variance distortion effects described in Proposition 2 when the resampling distribution is poorly aligned with the target allocation distribution for the rank-based statistic.

exhibits somewhat larger but still moderate errors together with power behavior similar to MC. The bootstrap approximation, by contrast, shows very large RMSE and effectively zero power at the nominal 0.05 level, indicating that it is not a reliable surrogate for the permutation Wilcoxon test under the simulation settings considered here. Unlike the t -statistic experiments, the BR supports used in this Wilcoxon study were not specifically optimized for the rank-sum statistic. Consequently, the Wilcoxon results are best interpreted as a robustness assessment showing that BR remains viable for a canonical rank-based test, rather than as a setting in which BR is expected to achieve maximal approximation accuracy relative to Monte Carlo permutation.

C.2 Reference interval comparisons

To complement the RMSE analysis in the main text, we compare benchmark p -values with nominal Monte Carlo uncertainty bands for the stochastic methods. These summaries are included only to provide a reference scale for stochastic simulation error and should not be interpreted as a separate inferential procedure for BR.

For a stochastic method producing an approximate p -value $\hat{p}_i$ from S resamples, define the nominal interval

$$\hat{p}_i \pm 1.96 \sqrt{\frac{\hat{p}_i(1 - \hat{p}_i)}{S}}.$$

We then report the containment proportion

$$\text{CP} = \frac{1}{M} \sum_{i=1}^M \mathbf{1} \left\{ p_i^{\text{exact}} \in \left[\hat{p}_i - 1.96 \sqrt{\frac{\hat{p}_i(1 - \hat{p}_i)}{S}}, \hat{p}_i + 1.96 \sqrt{\frac{\hat{p}_i(1 - \hat{p}_i)}{S}} \right] \right\},$$

where p_i^{exact} denotes the exact or high-precision benchmark p -value for replicate i .

For MC and BA, values of CP near 0.95 indicate behavior broadly consistent with nominal Monte Carlo uncertainty. For BR, these summaries have no confidence-interval interpretation, since BR is deterministic conditional on the support. They are reported only to provide a rough reference scale relative to the variability that would arise under stochastic resampling with the same nominal computational budget.

Table 9: Supplementary containment proportions CP across methods. For MC and BA, values near 0.95 indicate nominal Monte Carlo calibration. For BR, the values are reported only as reference benchmarks and are not confidence-interval coverage statements.

Method	Small N	Moderate N	Large N
MC	0.947	0.952	0.949
BA	0.934	0.928	0.921
BR	0.963	0.970	0.975

The BR entries in Table 9 should therefore be interpreted only as reference comparisons against the interval widths that would arise under stochastic resampling with the same nominal number of statistic evaluations.

C.3 Supplementary accuracy summaries

In addition to RMSE, we summarize the mean absolute error

$$\text{MAE} = \frac{1}{M} \sum_{i=1}^M |\hat{p}_i - p_i^{\text{exact}}|$$

and the decision agreement rate

$$\text{AR}_\alpha = \frac{1}{M} \sum_{i=1}^M \mathbf{1} \{ \mathbf{1}(\hat{p}_i \leq \alpha) = \mathbf{1}(p_i^{\text{exact}} \leq \alpha) \},$$

evaluated at the nominal level $\alpha = 0.05$.

Table 10: Supplementary accuracy summaries. MAE is mean absolute error relative to the exact or high-precision benchmark; $\text{AR}_{0.05}$ is the agreement rate for reject/non-reject decisions at level 0.05. Lower MAE and higher agreement are preferable.

Method	MAE	$\text{AR}_{0.05}$
MC	0.017	0.931
BA	0.031	0.882
BR	0.011	0.957

These summaries complement RMSE by separating pointwise approximation error from agreement in testing decisions. They are descriptive and should be interpreted together with the main simulation results rather than as stand-alone inferential claims.

C.4 Distribution-specific behavior

The simulation study includes symmetric and skewed data-generating distributions. To illustrate heterogeneity across settings, Table 11 reports representative RMSE summaries by distribution.

Table 11: Supplementary distribution-specific RMSE summaries. Values are averaged over sample-size regimes and hypotheses considered in the simulation study. Lower values indicate more accurate approximation to the exact permutation benchmark.

Distribution	BR	MC	BA
Normal	0.012	0.018	0.031
Uniform	0.011	0.017	0.029
Exponential	0.014	0.024	0.046
Gamma	0.015	0.026	0.049
Lognormal	0.016	0.028	0.053

The qualitative pattern in Table 11 is consistent with the main text: BR performs competitively in symmetric settings and tends to show larger gains in skewed settings, where stochastic approximation error is more pronounced under fixed resampling budgets.

C.5 Supplementary runtime summaries

The reported timings were obtained in a fixed computing environment using the software implementation described in the main text. They are reported to document relative computational scaling across methods rather than machine-independent absolute runtimes.

To complement the runtime analysis in the main text, we report representative average runtimes across broad sample-size regimes.

Table 12: Supplementary average runtime summaries in seconds across sample-size regimes. These values are representative of the study environment and should be interpreted comparatively rather than as machine-independent constants.

Method	Small N	Moderate N	Large N
FP (exact)	> 300	Infeasible [†]	Infeasible [†]
MC ($S = 10^4$)	12.40	18.70	31.20
BA ($S = 10^4$)	14.10	21.50	35.60
BR ($ \mathcal{B}_R \approx 300$)	0.92	1.31	1.87

[†]Exact FP requires enumeration over all $\binom{N}{n}$ treatment allocations, which grows combinatorially with N and becomes computationally impractical in the moderate and large regimes considered here.

These timings are reported to document relative computational scaling in the study environment. They should not be interpreted as universal runtime guarantees, since actual performance depends on implementation details, hardware, and the cost of evaluating the chosen test statistic.

C.6 High-dimensional illustration: supplementary summaries

For the high-dimensional genomic-style example, we summarize agreement with the benchmark permutation procedure at both the raw-significance and multiplicity-adjusted levels. Let

$$\mathcal{R}_{\text{meth}} = \{j : \hat{p}_{j,\text{meth}} \leq \alpha\}$$

denote the rejection set for a given method at level α , and let $\mathcal{R}_{\text{exact}}$ denote the benchmark rejection set.

We report:

1. the raw agreement rate

$$\frac{1}{p} \sum_{j=1}^p \mathbf{1} \{ \mathbf{1}(\hat{p}_{j,\text{meth}} \leq \alpha) = \mathbf{1}(\hat{p}_{j,\text{exact}} \leq \alpha) \};$$

2. the Jaccard index

$$J(\mathcal{R}_{\text{meth}}, \mathcal{R}_{\text{exact}}) = \frac{|\mathcal{R}_{\text{meth}} \cap \mathcal{R}_{\text{exact}}|}{|\mathcal{R}_{\text{meth}} \cup \mathcal{R}_{\text{exact}}|};$$

3. the corresponding summaries after Benjamini–Hochberg adjustment.

These summaries are included to document the practical behavior of BR in a large-scale testing setting. They are supplementary empirical evidence and are not a substitute for a separate multiple-testing theory for BR-based procedures.

Table 13: Supplementary summaries for the high-dimensional example. Agreement and Jaccard similarity are reported relative to a high-precision permutation benchmark.

Method	Raw agreement	Raw Jaccard	BH agreement	BH Jaccard
MC	0.944	0.811	0.951	0.826
BA	0.901	0.734	0.918	0.759
BR	0.963	0.846	0.968	0.857

C.7 Interpretation

The numerical summaries in this appendix are intended to complement the theoretical results and the main simulation tables. They support three broad empirical observations:

1. near-balanced BR supports can be constructed with small low-order discrepancy;
2. BR can closely approximate exact or high-precision permutation benchmarks while remaining deterministic;
3. BR can offer substantial computational savings relative to stochastic resampling under the study settings considered here.

These observations should be interpreted as empirical evidence from the present simulation design, not as universal guarantees across all statistics, sample sizes, or resampling regimes. They are intended to reinforce, rather than modify, the principal conclusions drawn in the main text.

C.8 Supplementary orbit summaries for $N = 7$, $n = 3$

Table 14 summarizes, for the $N = 7$, $n = 3$ example in Section 2.3, the BR mean and variance of T , the resulting BR p-value, and the combinatorial class of each seed block. The full permutation benchmark is included in the last row for reference.

Table 14: Representative BR summaries for seed-specific orbits in the $N = 7$, $n = 3$ example. The final row reports the full-permutation benchmark over all $\binom{7}{3} = 35$ allocations.

Seed ($\mathcal{D}$)	Combinatorial class	$\mathbb{E}_{\text{BR}}[T]$	$\text{Var}_{\text{BR}}(T)$	p_{BR}	Agreement at $\alpha = 0.05$
$A_1 = \{1, 2, 4\}$	Cyclic difference set	2.8428	1.7042	0.2857	Fail to reject
$A_2 = \{1, 3, 5\}$	Asymmetric sub-orbit	2.5457	3.2048	0.5714	Fail to reject
$A_3 = \{1, 6, 7\}$	Single-gap cluster	2.4086	1.4112	1.0000	Fail to reject
$A_4 = \{2, 3, 6\}$	Single-gap cluster	2.4086	1.4112	0.5714	Fail to reject
$A_5 = \{2, 5, 7\}$	Asymmetric sub-orbit	2.5457	3.2048	0.7143	Fail to reject
$A_6 = \{3, 4, 7\}$	Single-gap cluster	2.4086	1.4112	0.8571	Fail to reject
$A_7 = \{4, 5, 6\}$	Single-gap cluster	2.4086	1.4112	0.2857	Fail to reject
Full FP	All $\mathcal{A}_{7,3}$	2.8428	1.7042	0.3143	Reference

These summaries illustrate that only the cyclic difference-set seed reproduces the full allocation mean and variance for T , whereas other seeds exhibit variance distortion consistent with Proposition 2. Nonetheless, for this particular dataset, all seed-specific BR procedures yield the same inferential decision at level $\alpha = 0.05$ as the exact permutation test.

Conflict of interest

The authors declare no conflict of interest.

References

- [1] Besag, J. and Clifford, P. (1991). Sequential Monte Carlo p-values. *Biometrika*, 78(2), 301–304. doi:[10.1093/biomet/78.2.301](https://doi.org/10.1093/biomet/78.2.301)
- [2] Chung, E. and Romano, J. P. (2013). Multivariate and multiple permutation tests. *Journal of Econometrics*, 173(1), 76–91. doi:[10.1016/j.jeconom.2012.10.002](https://doi.org/10.1016/j.jeconom.2012.10.002)
- [3] Efron, B. and Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall, New York. doi:[10.1201/9780429246593](https://doi.org/10.1201/9780429246593)
- [4] Fisher, R. A. (1935). The logic of inductive inference. *Journal of the Royal Statistical Society*, 98(1), 39–54. doi:[10.2307/2342435](https://doi.org/10.2307/2342435)
- [5] Gandy, A. (2009). Sequential implementation of Monte Carlo tests with uniformly bounded resampling risk. *Journal of the American Statistical Association*, 104(488), 1504–1511. doi:[10.1198/jasa.2009.tm08368](https://doi.org/10.1198/jasa.2009.tm08368)
- [6] Hemerik, J., Thoresen, M. and Finos, L. (2021). Permutation testing in high-dimensional linear models. *Journal of Statistical Computation and Simulation*, 91(7), 1403–1421. doi:[10.1080/00949655.2020.1836183](https://doi.org/10.1080/00949655.2020.1836183)
- [7] Hemerik, J. and Goeman, J. J. (2018). Exact testing with random permutations. *TEST*, 27(4), 811–830. doi:[10.1007/s11749-017-0571-1](https://doi.org/10.1007/s11749-017-0571-1)
- [8] Hoeffding, W. (1951). A combinatorial central limit theorem. *Annals of Mathematical Statistics*, 22(4), 558–566. doi:[10.1214/aoms/1177729545](https://doi.org/10.1214/aoms/1177729545)
- [9] Hoeffding, W. (1952). The large-sample power of tests based on permutations of observations. *The Annals of Mathematical Statistics*, 23(2), 169–192. doi:[10.1214/aoms/1177729438](https://doi.org/10.1214/aoms/1177729438)
- [10] Janssen, A. (2003). How do bootstrap and permutation tests work? *The Annals of Statistics*, 31(3), 768–806. doi:[10.1214/aos/1056562461](https://doi.org/10.1214/aos/1056562461)
- [11] Lehmann, E. L. and Romano, J. P. (2005). *Testing Statistical Hypotheses*, 3rd edn. Springer, New York. doi:[10.1007/0-387-27605-X](https://doi.org/10.1007/0-387-27605-X)
- [12] Mammen, E. (1993). Bootstrap and wild bootstrap for high dimensional linear models. *The Annals of Statistics*, 21(1), 255–281. doi:[10.1214/aos/1176349025](https://doi.org/10.1214/aos/1176349025)
- [13] Niederreiter, H. (1992). *Random Number Generation and Quasi-Monte Carlo Methods*. SIAM, Philadelphia.
- [14] Pesarin, F. and Salmaso, L. (2010). *Permutation Tests for Complex Data: Theory, Applications and Software*. Wiley, Chichester. doi:[10.1002/9780470689516](https://doi.org/10.1002/9780470689516)
- [15] Phipson, B. and Smyth, G. K. (2010). Permutation p-values should never be zero: calculating exact p-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1), Article 39. doi:[10.2202/1544-6115.1585](https://doi.org/10.2202/1544-6115.1585)

- [16] Politis, D. N., Romano, J. P. and Wolf, M. (1999). *Subsampling*. Springer, New York. doi:[10.1007/978-1-4612-1554-7](https://doi.org/10.1007/978-1-4612-1554-7)
- [17] Romano, J. P. and Wolf, M. (2005a). Exact and approximate stepdown methods for multiple hypothesis testing. *Journal of the American Statistical Association*, 100(469), 94–108. doi:[10.1198/016214504000000539](https://doi.org/10.1198/016214504000000539)
- [18] Romano, J. P. and Wolf, M. (2005b). Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4), 1237–1282. doi:[10.1111/j.1468-0262.2005.00615.x](https://doi.org/10.1111/j.1468-0262.2005.00615.x)
- [19] Southworth, H., Pesarin, F. and Smyth, G. K. (2009). Permutation p-values for complex-valued statistics with application to testing for equivalence in two-group designs. *Biostatistics*, 10(1), 200–211. doi:[10.1093/biostatistics/kxn031](https://doi.org/10.1093/biostatistics/kxn031)
- [20] Stinson, D. R. (2004). *Combinatorial Designs: Constructions and Analysis*. Springer, New York.
- [21] Wald, A. and Wolfowitz, J. (1944). Statistical tests based on permutations of the observations. *Annals of Mathematical Statistics*, 15(4), 358–372. doi:[10.1214/aoms/1177731207](https://doi.org/10.1214/aoms/1177731207)
- [22] Wu, C. F. J. (1986). Jackknife, bootstrap and other resampling methods in regression analysis. *The Annals of Statistics*, 14(4), 1261–1295. doi:[10.1214/aos/1176350142](https://doi.org/10.1214/aos/1176350142)
- [23] Kim, S., Zhao, Z. and Shao, X. (2015). Nonparametric functional central limit theorem for time series regression with application to self-normalized confidence intervals. *Journal of Multivariate Analysis*, 133, 277–290. doi:[10.1016/j.jmva.2014.09.017](https://doi.org/10.1016/j.jmva.2014.09.017)